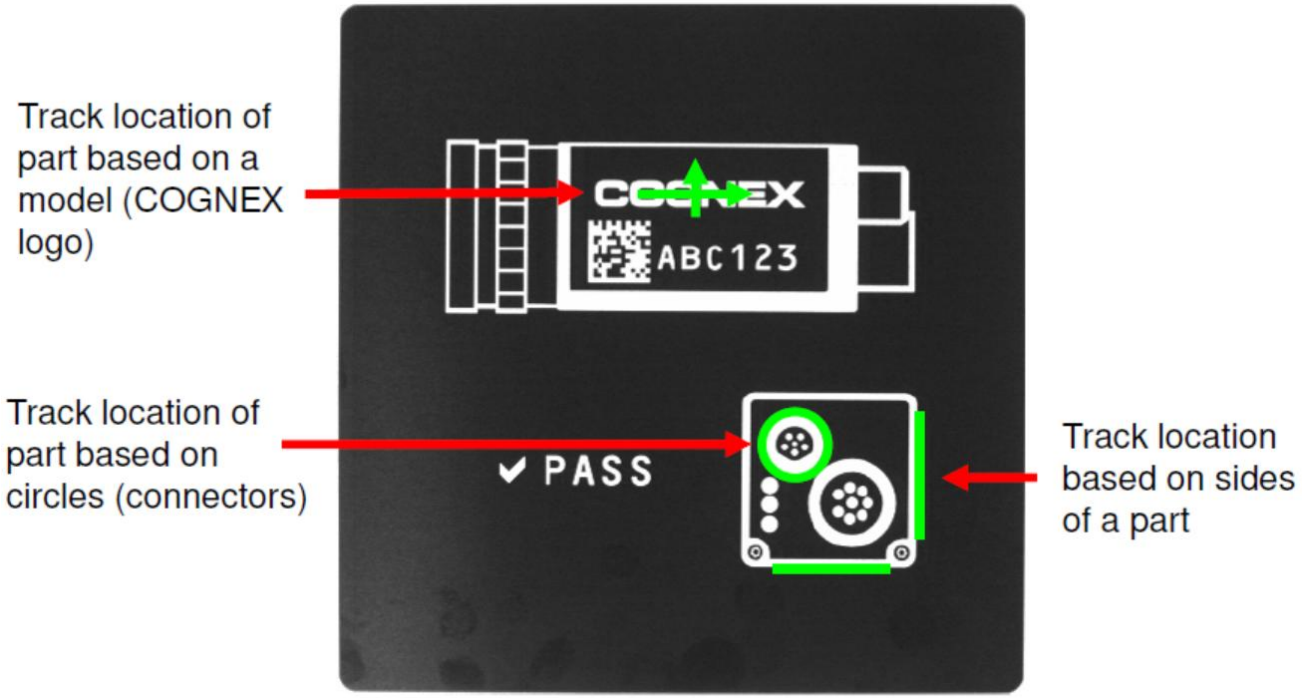
Inspection Tools & Lighting

Lecture 4

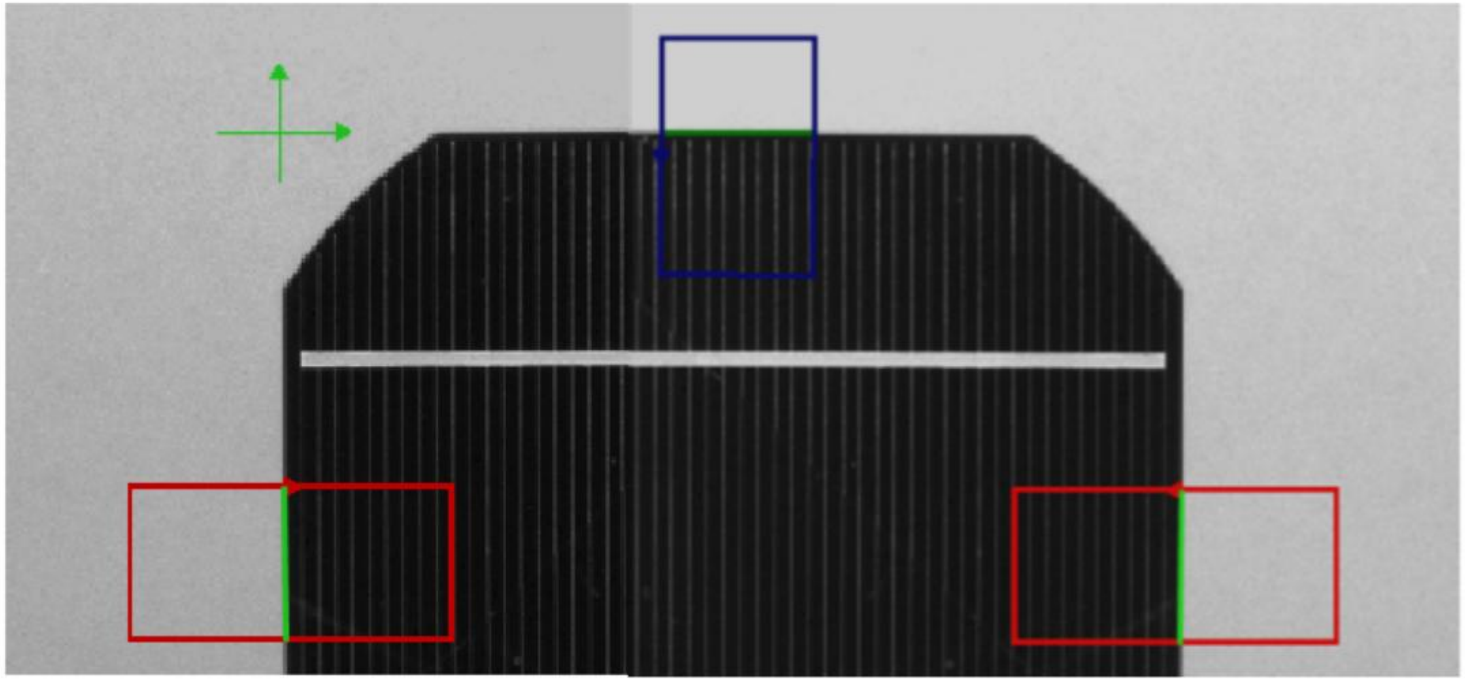
Ways to Locate a Part

- What other **Location Tools** can you use on this part?
  - Track the location of the part based on the **circles**.
  - Track the location of the part based on the **sides of the part**.
  - How do these methods compare with using a **Pattern tool**?
  - **Pattern tools** can tolerate **changes** in the position and orientation of a part.
  - **Circle and Edge tools** require the selected feature to remain relatively **fixed** in position and orientation.



Edge Tools

- The **Edge Tools** detect edges by examining a **region of interest (ROI)** for discontinuities in pixel polarity — an area in the image where there is an **abrupt change in intensity** between light and dark pixels.
- An edge may span only **two or three pixels**, or it may extend across **several pixels**.



Edge Tools

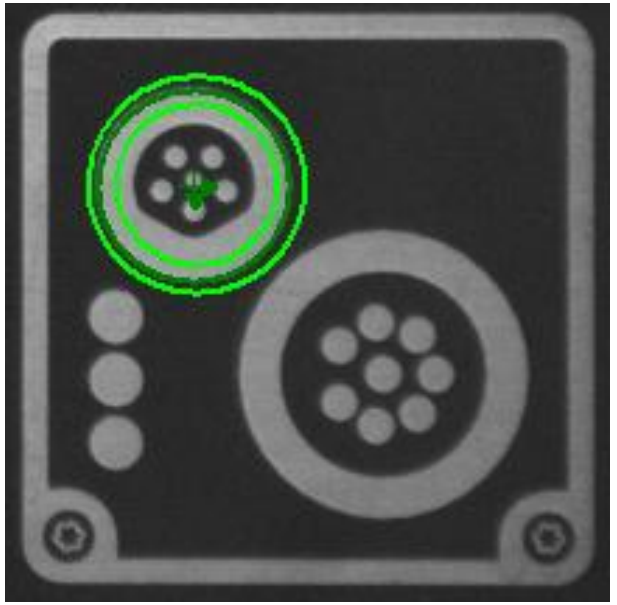
- The **Edge Tools** should be used in the following circumstances:
  - The edge has **high contrast** between light and dark pixels.
  - The application requires **quick detection of features**. The Edge Tools are among the fastest In-Sight Vision Tools, providing faster detection of edge features than Pattern Tools.
  - Edge Tools only analyze **greyscale images**; all color images are automatically converted to greyscale.

Edge Tools

- **Edge Detection** can be extremely useful in applications such as the following:
  - **Gauging applications**, such as measuring the radius of circular features on parts or the length of circuits on a circuit board.
  - **Locating features on parts**, such as circular features or the straight ends of parts, to use as fixture inputs for other In-Sight Vision Tools.
  - **Presence/absence applications**, where the Edge Tools are used to determine whether an edge feature is present or absent.
  - **Counting and sorting applications**, where the Edge Tools are used to count and sort edges based on their established parameters.

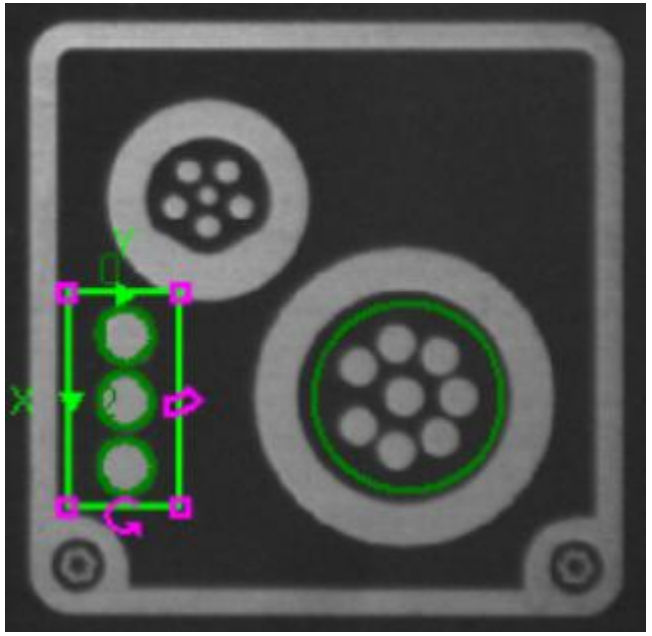
Circle Tools

- The **Circle Tools** are used to detect and locate **circular features** within a specified ROI. It identifies the circle and can report such as its **X and Y coordinates** (center) and **diameter**.
- The **Circle Tools** should be used in the following circumstances:
  - The circular feature has **high contrast** with the surrounding area.
  - The application requires **quick detection** of circular features. The Circle Tools are one of the faster In-Sight Vision Tools.



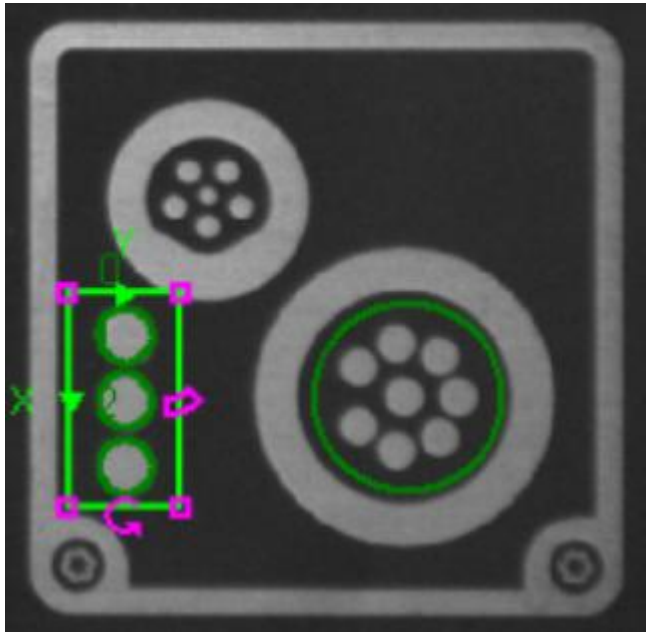
Blob Tools

- In machine vision, any **two-dimensional, variably shaped, similarly shaded group of connected pixels** can be referred to as a **blob**.
- For example, a **light-colored shape on a dark background**.
- The process of **extracting blobs from an image, identifying them within an ROI, and sorting them based on specific criteria** is called **blob analysis**.



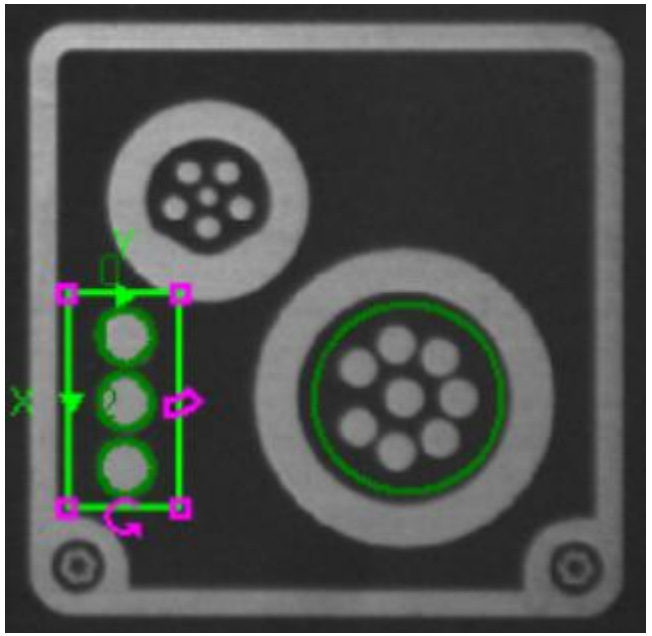
Blob Tools

- During this process, the pixels in the ROI are divided into two categories, **Blob** and **Background**, based on the **greyscale threshold value**.
- The threshold value separates the pixels into two categories: pixels **below the threshold** are classified as **Black**, while pixels **above the threshold** are classified as **White**.



Blob Tools

- The **Blob Tool** should be used in the following circumstances:
  - The image contains a **two-dimensional object**.
  - The image provides **high contrast between light and dark regions**.



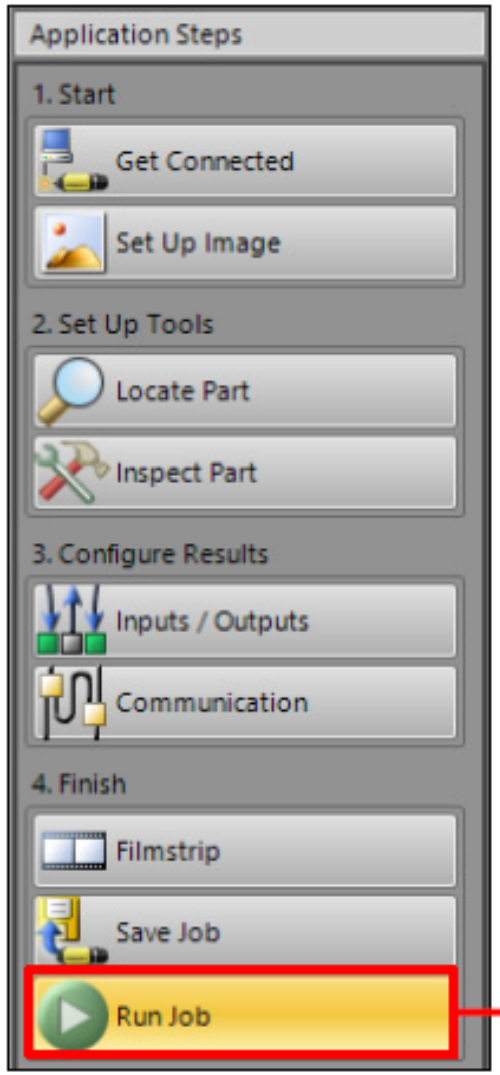


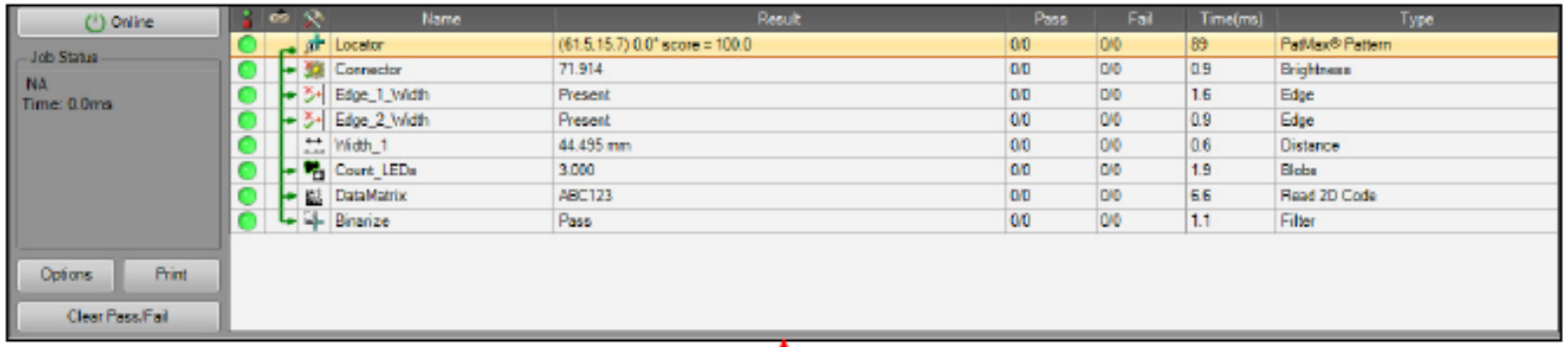


Blob Tools

- **Blob Analysis** can be extremely useful in the following applications:
  - **Coarse gauging applications**, such as measuring the diameter of circular features or the dimensions of squares to determine a part's width in a particular orientation.
  - **First-pass (initial) positioning applications**, where the Blob Tool is used to quickly locate and establish a rough fixture for another tool.
  - **Presence/absence applications**, where the Blob Tool is used to determine whether a blob is present or absent in the image.
  - **Counting applications**, where the Blob Tool can be used to count and sort blobs based on characteristics such as **area or perimeter**.

Run Job


Application Steps
1. Start
Get Connected
Set Up Image
2. Set Up Tools
Locate Part
Inspect Part
3. Configure Results
Inputs / Outputs
Communication
4. Finish
Filmstrip
Save Job
Run Job


Online
Job Status
NA
Time: 0.0ms
Options
Print
Clear Pass/Fail
Name
Result
Pass
Fail
Time(ms)
Type
Locator
(61.5,15.7) 0.0° score = 100.0
0/0
0/0
89
PatMax® Pattern
Connector
71.914
0/0
0/0
0.9
Brightness
Edge_1_Width
Present
0/0
0/0
1.6
Edge
Edge_2_Width
Present
0/0
0/0
0.9
Edge
Width_1
44.495 mm
0/0
0/0
0.6
Distance
Count_LEDs
3.000
0/0
0/0
1.9
Blobs
DataMatrix
ABC123
0/0
0/0
6.6
Read 2D Code
Binarize
Pass
0/0
0/0
1.1
Filter

Run Job

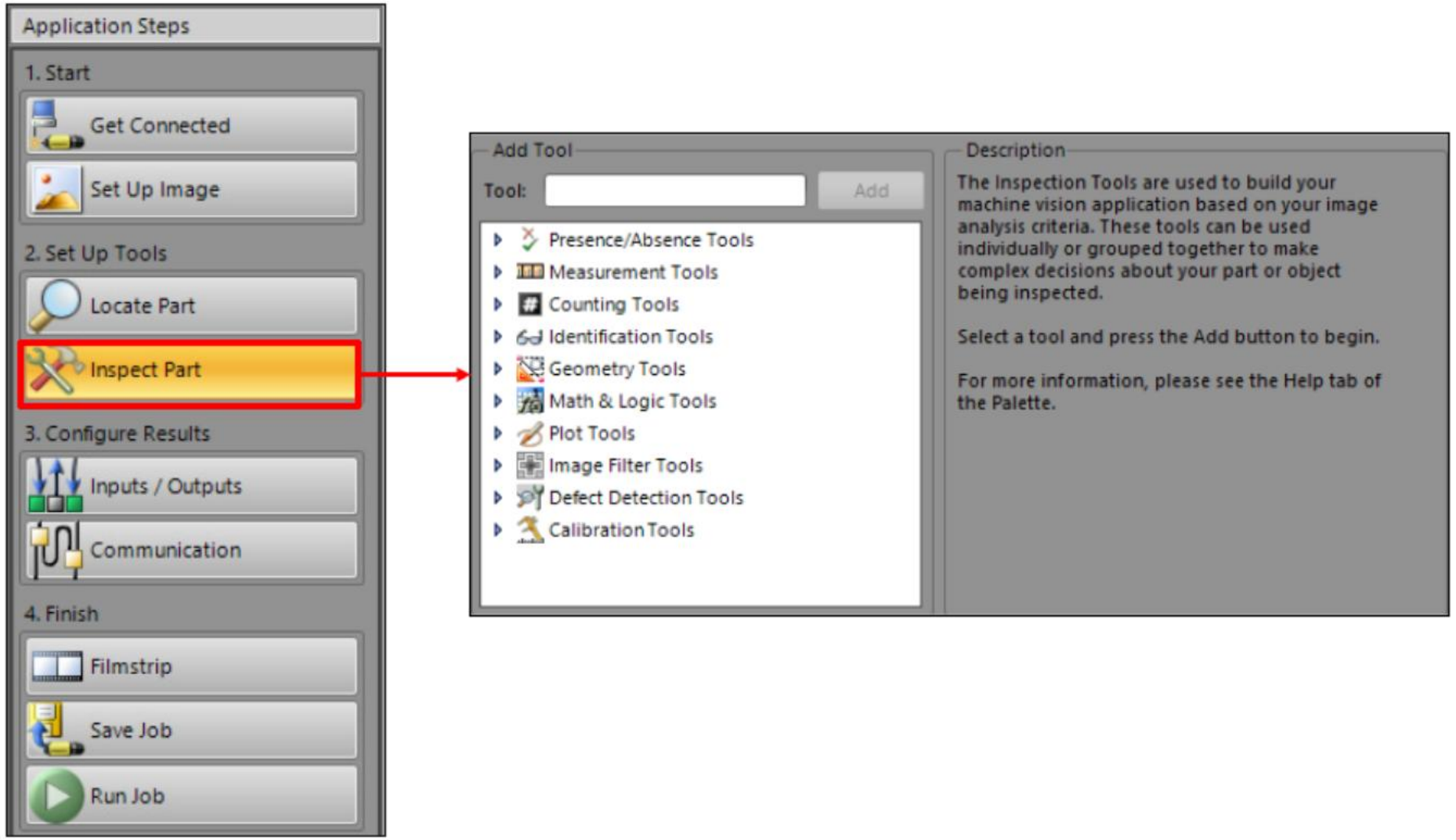
- Once all job parameters are set, you can run the job, review the results, and modify or troubleshoot the settings. You can test the job to ensure that all parameters are within their specified limits and that the In-Sight vision system is accurately acquiring images.
- **Running a job on a camera** simply involves placing the camera **Online**.
- **To place the camera Online:** Click the **Online** button in the Settings pane. The button is a toggle and turns **orange** to indicate its Online status.
- **To place the camera Offline:** Click the **orange Online** button in the Settings pane. The button reverts to its original color to indicate its Offline status.

Run Job

- The **Results Table**, viewable in the **Run Job** step in EasyBuilder, displays information about each tool in the active job, allowing you to monitor and troubleshoot the tools as the job runs.
- For each tool, the Results Table displays its **pass/fail status, tool icon, name, result, pass total, fail total**, and the **time, in milliseconds, required to run the tool**.

Inspection Tools

Inspection Part



What Are Inspection Tools?

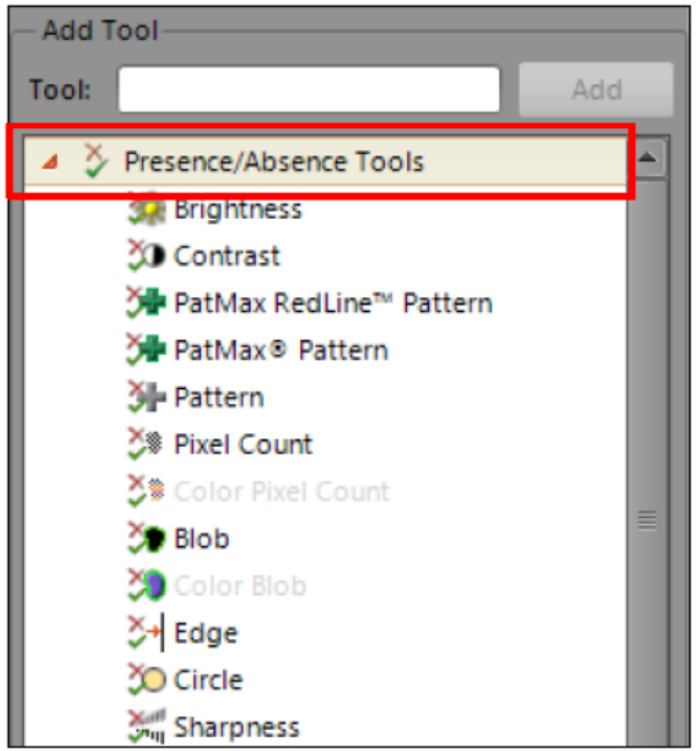
- **Inspection Tools** are used to create a machine vision application based on the features you want to analyze.
- Tools can be used individually or **combined to make more complex decisions** about the part or object being inspected.
- In EasyBuilder, tools are organized by their **function or task**, rather than by the **tool type** itself.

Inspection Tools

- **Presence/Absence –** Checks whether a feature is present.
- **Measurement –** Gauges and measures features.
- **Counting –** Determines the number of instances of a feature or part.
- **Identification –** Finds barcodes, 2D codes, OCV/OCR, patterns, ID, and color ID.
- **Geometry –** Creates lines, points, and angles.
- **Math & Logic –** Performs mathematical and logical operations and trend analysis.
- **Plot –** Draws text and shapes on the image.
- **Image Filter –** Alters the image to enhance or manipulate features.
- **Defect Detection –** Finds defects on edges and beads.
- **Calibration –** Provides advanced calibration functions.

Presence/Absence Tools

- The **Presence/Absence** Tools are used to determine whether a feature is present or absent in the image.





Presence/Absence Tools

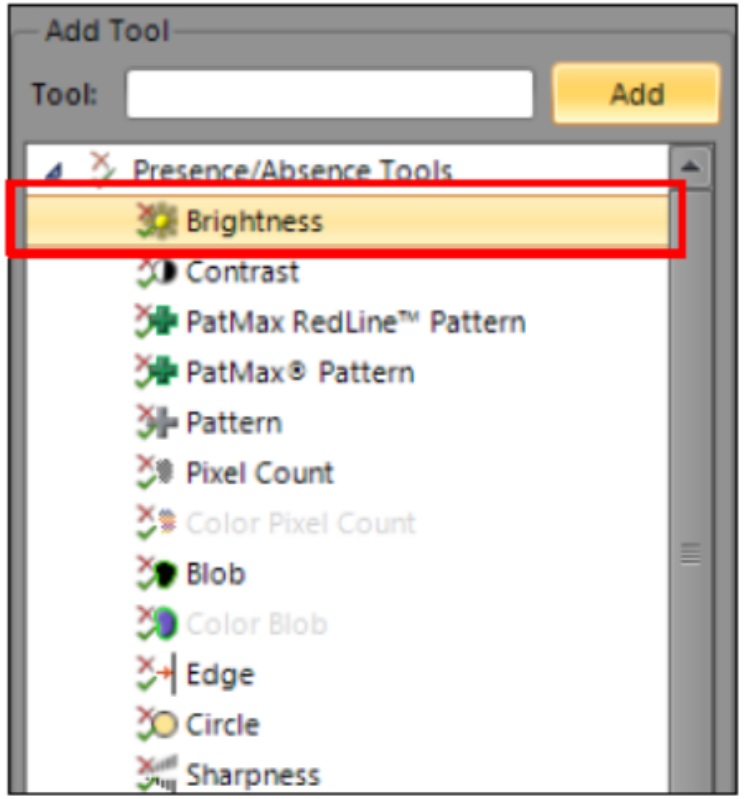
- How do the tools determine whether a feature is **present or absent** in the image?
  - **Brightness** – Examines the **average greyscale value** of the pixels within the region.
  - **Contrast** – Examines the **difference between the mean greyscale value above the threshold and the mean greyscale value below the threshold**.
  - **PatMax Pattern** – Uses the **PatMax algorithm** to locate a feature based on a trained representation of the pattern.
  - **Pattern** – Looks for a pattern based on a **trained representation** of that pattern.

Presence/Absence Tools

- How do the tools determine whether a feature is **present or absent** in the image?
  - **Pixel Count –** Determines the result based on the number of **white or black pixels** in the region.
  - **Color Pixel Count –** Determines the result based on the number of **pixels of a specified color** in the region.
  - **Blob –** Determines whether a **blob feature** is present or absent.
  - **Color Blob –** Determines whether a **color blob feature** is present or absent.
  - **Edge –** Looks for **linear edge features** that fall within the specified parameters.
  - **Circle –** Looks for **circular edge features** that fall within the specified parameters.
  - **Sharpness –** Determines whether an image is **properly focused** based on the **image sharpness score** within the region.

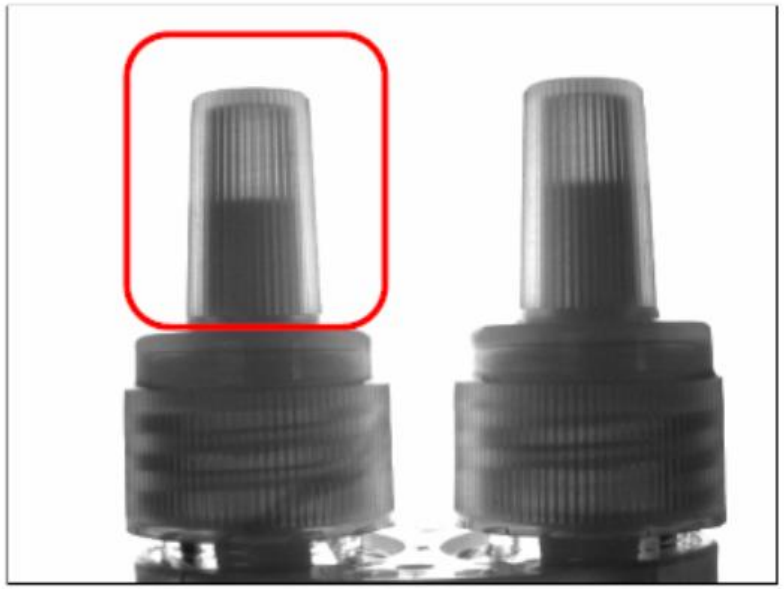
Brightness Tool

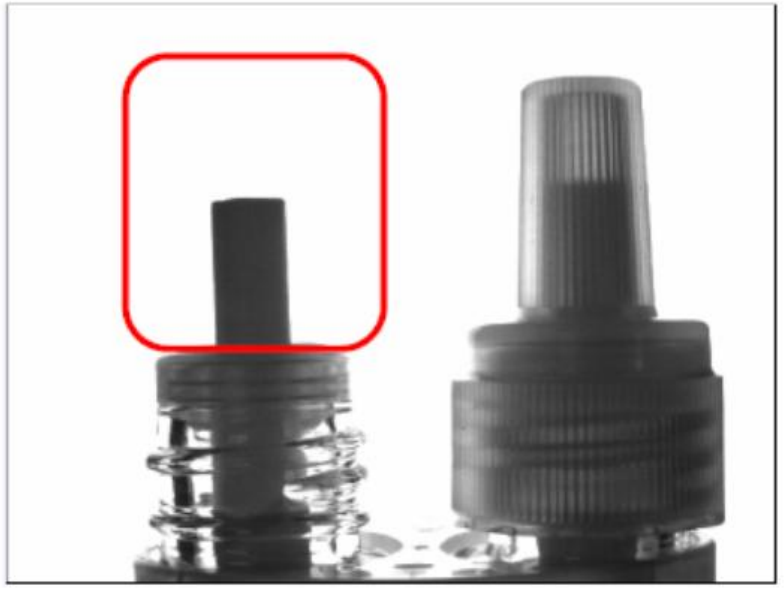
- The **Brightness Tool** determines whether a feature is present or absent based on the **average greyscale value (brightness)** of the pixels within the region and reports that value.
- The **Brightness Tool** is useful when the feature on a **good part** is significantly darker or lighter than the corresponding feature on a **bad part**.



Brightness Tool

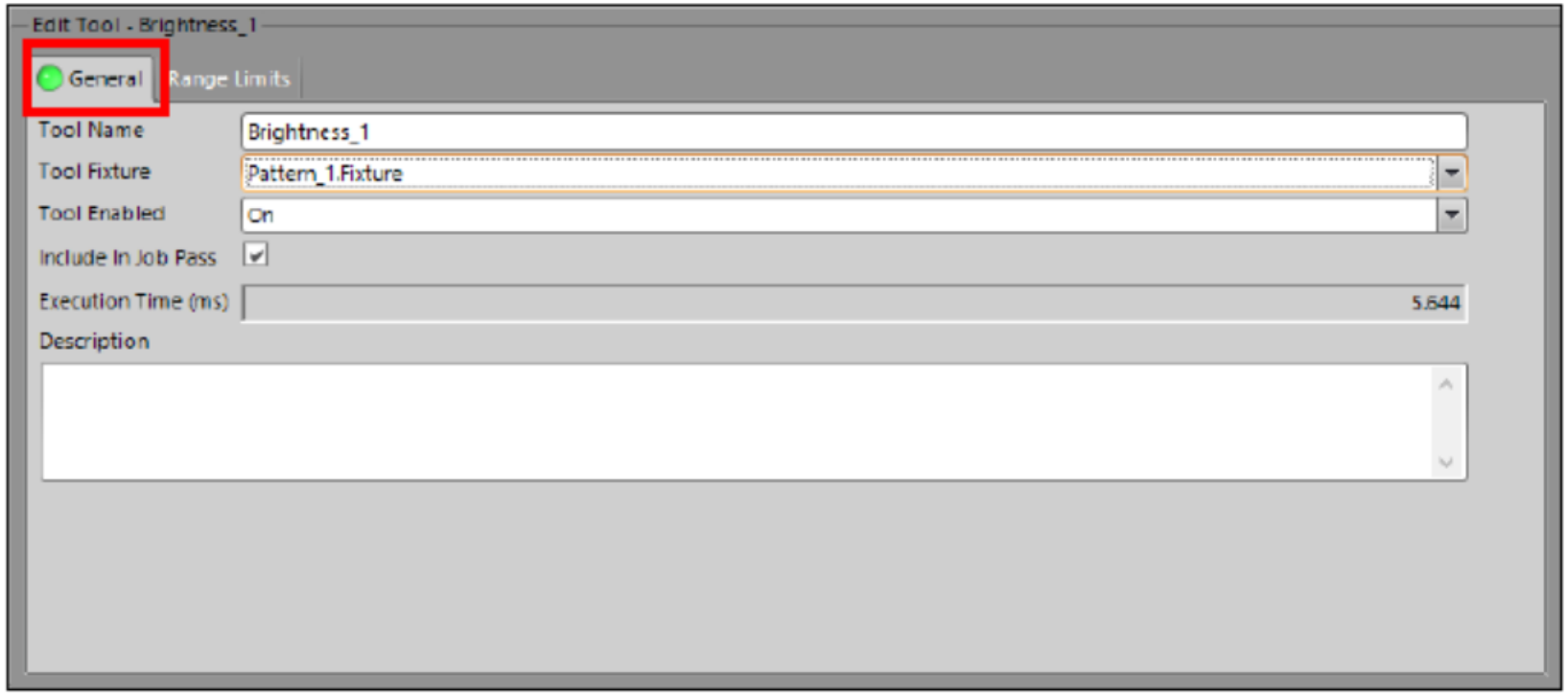
- For example, the presence or absence of a **dark- or light-colored bottle cap** can be quickly determined by measuring the brightness within the region.





Brightness Tool

- After the tool's region has been set and accepted, you can begin customizing and fine-tuning the tool for your application. The **Settings pane** of the Brightness Tool consists of two tabs:



Brightness Tool

General Tab

- **Tool Name** – Defines the name of the inspection tool, which is used when configuring the results (**Inputs, Outputs, and Communication**). The default name is **Brightness_1**. As additional Brightness Tools are added, the number at the end increases. You can customize the tool by giving it a name associated with your application.
- **Tool Fixture** – Defines a fixture for the tool. This control is enabled only if another tool that defines a fixture has already been added. **It defaults to the most recent successful Locator Tool.**

Brightness Tool

General Tab

- **Tool Enabled** – Defines when and whether the inspection tool should run. The default setting is **On**. Selecting **Off** disables the tool. Selecting **Input** ties the tool to a specific discrete input line, allowing an external control system, such as a **PLC**, to turn the tool on or off as needed.
- **Execution Time** – Displays the amount of time, in milliseconds, that the tool takes to perform its inspection.

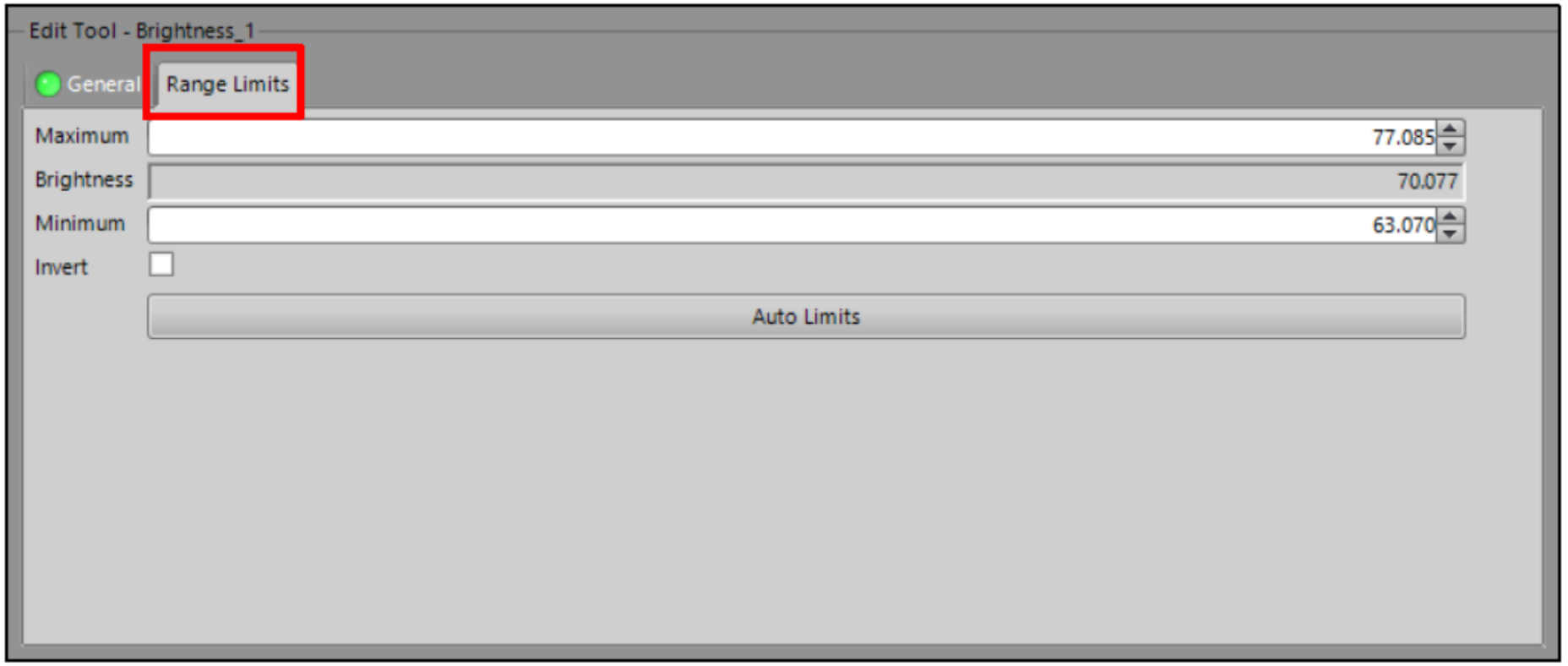
Brightness Tool

General Tab

- **Include in Job Pass** – Determines whether a tool's pass/fail result affects the overall job result. By default, this option is checked, meaning the tool's result is included in the overall job result. Uncheck it to keep the tool's result separate from the overall job result, which is useful when a tool is expected to fail without affecting the overall job outcome.
- **Description** – Defines a text message that can be associated with the tool. By default, this field is blank.

Brightness Tool

- **Range Limits Tab**



Brightness Tool

Range Limits Tab

- **Maximum / Minimum** – Defines the **maximum and minimum acceptable average greyscale values** (0 to 255). The tool automatically generates a default **Smart Limit** based on the image, but you can adjust these values to match your application's requirements.
- **Brightness** – Reports the **average greyscale value** of the pixels within the region (0 to 255). Values that fall **at or between** the Minimum and Maximum settings are classified as **Pass**.

Brightness Tool

Range Limits Tab

- **Invert Results** – Defines whether the **Pass/Fail logic** of the tool should be inverted (default = **Off**, unchecked).
  - **Unchecked** – Standard results are applied. The tool result is **Pass** if the feature is detected and falls within the **Range Limits**.
  - **Checked** – The results are inverted. The tool result is **Fail** if the feature is detected and falls within the **Range Limits**.

Brightness Tool

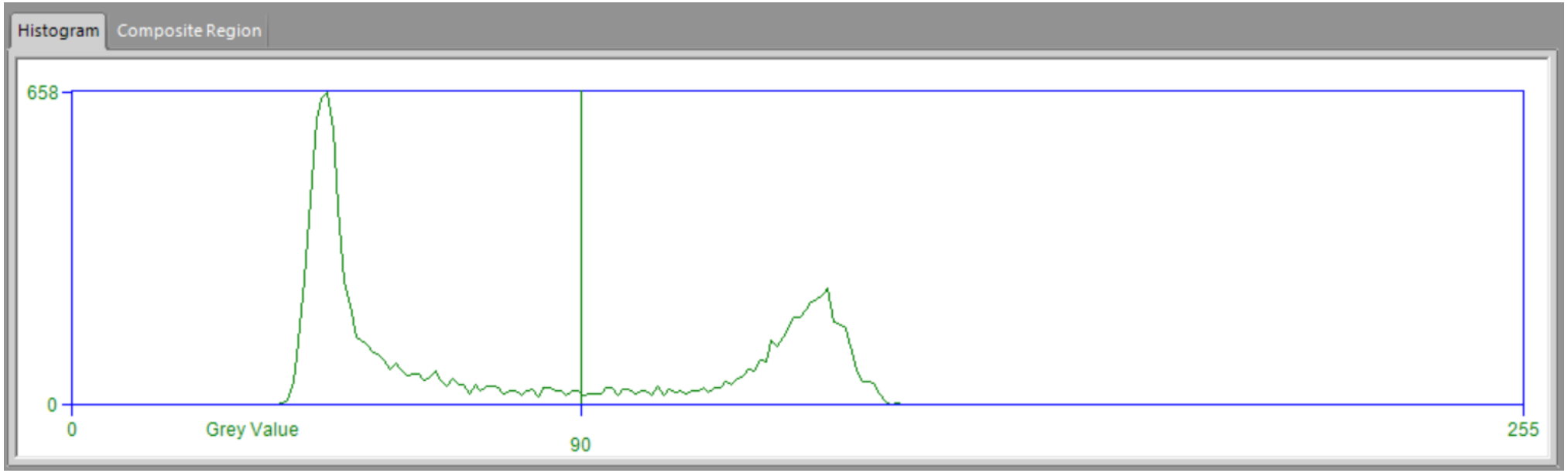
Range Limits Tab

- **Auto Limits Button** – Defines the limits based on the **Smart Limits** settings. This button is useful when a parameter has been changed and the limits need to be adjusted, or when a tool has been added and no limits have been defined.
- When a tool with a **Range Limits** tab is added to your job, the tool automatically examines the image and determines appropriate **Minimum and Maximum limits** based on the requirements of the tool.

Brightness Tool

Histogram Tab

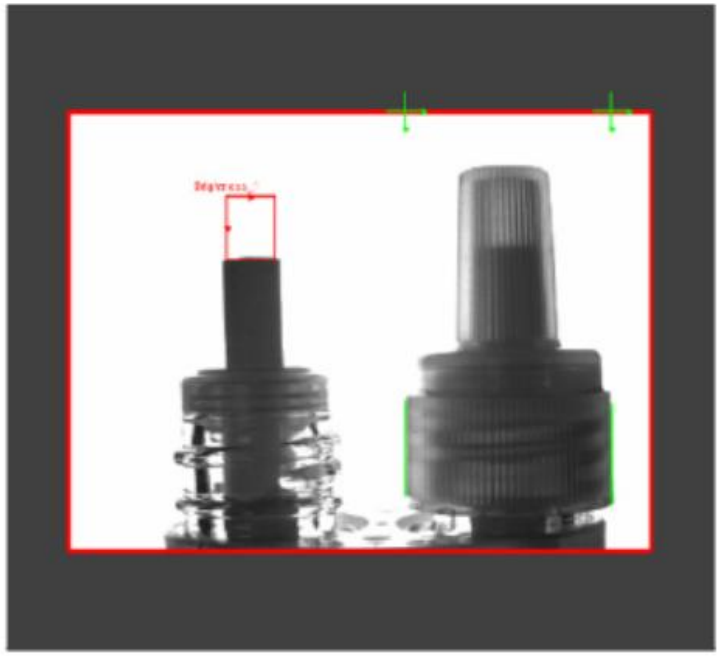
- The **X-axis** of the graph represents the **greyscale values**, ranging from **0 to 255**.
- The **Y-axis** represents the **number of pixels** at each greyscale value. The scale is established based on the greyscale value with the **greatest number of pixels**.
- The **green vertical line** indicates the **threshold value**.



Brightness Tool

- The first image has **low brightness**, indicating that the **bottle cap is present**. Therefore, image one represents a **good part** and will **Pass**.
- The second image has **high brightness**, indicating that the **bottle cap is missing**. Therefore, image two represents a **bad part** and will **Fail**.



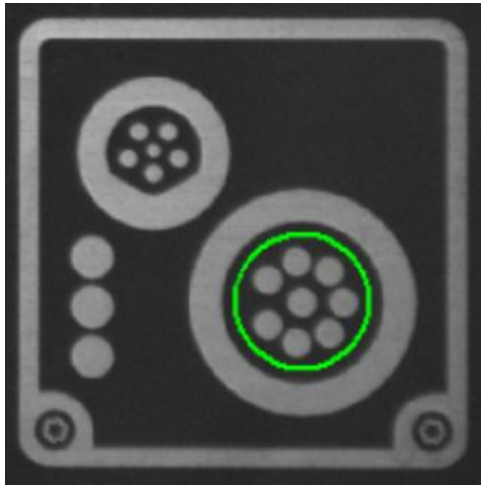


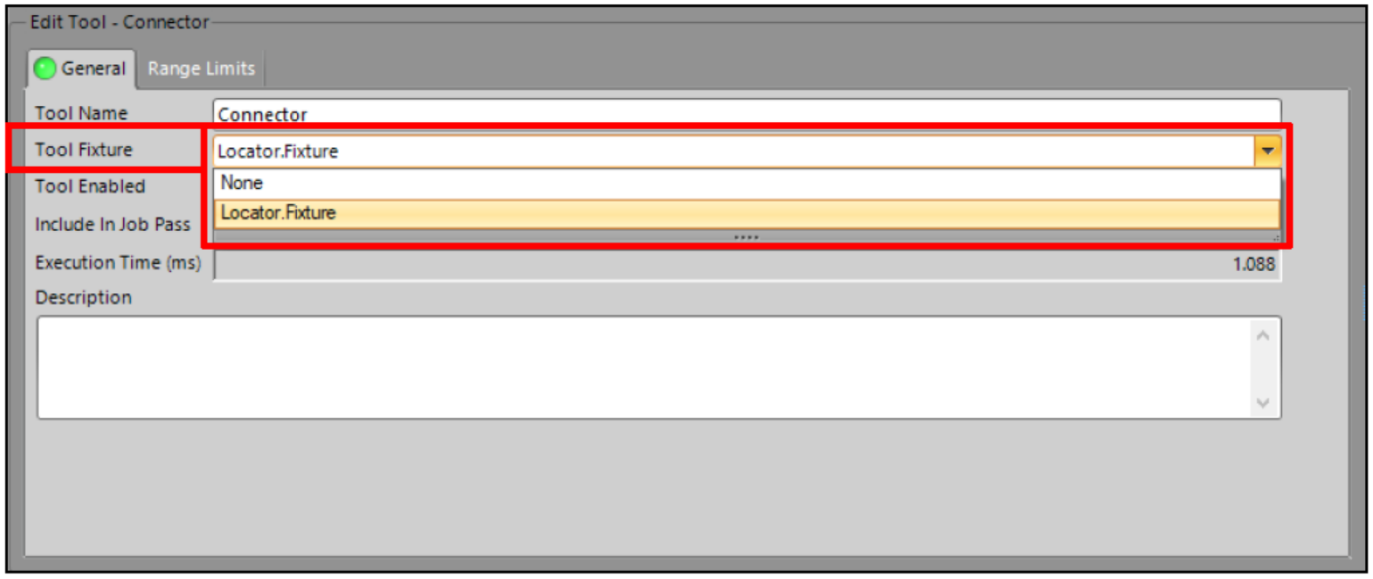
Guidelines for Choosing A Feature

- When choosing a feature for your inspection, consider the following guidelines to ensure the best results:
  - If you are not using a fixture, ensure that the feature is located in **roughly the same position** in every image, or that the **region is large enough to accommodate the expected movement** of the feature.
  - The **Brightness Tool** is useful when the feature being inspected has a **consistent greyscale value when present** (for example, black) and a **different greyscale value when absent**.

Fixture to A Location Tool

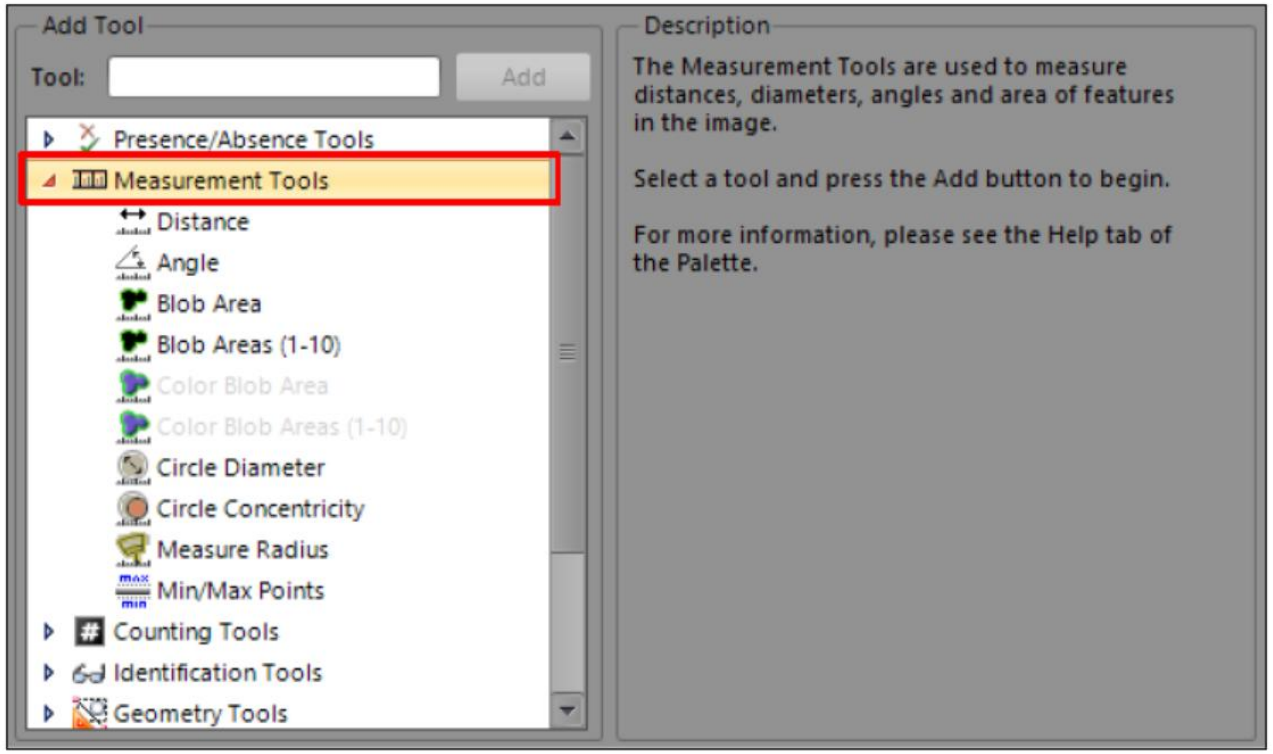
- Once you have set up one or more **Location Tools**, EasyBuilder allows you to select which **Location Tool** should be used to fixture each inspection tool.
- You can select **None** if you do not want to fixture the inspection tool.
- EasyBuilder automatically applies the selected **Location Tool** to fixture the inspection tool.





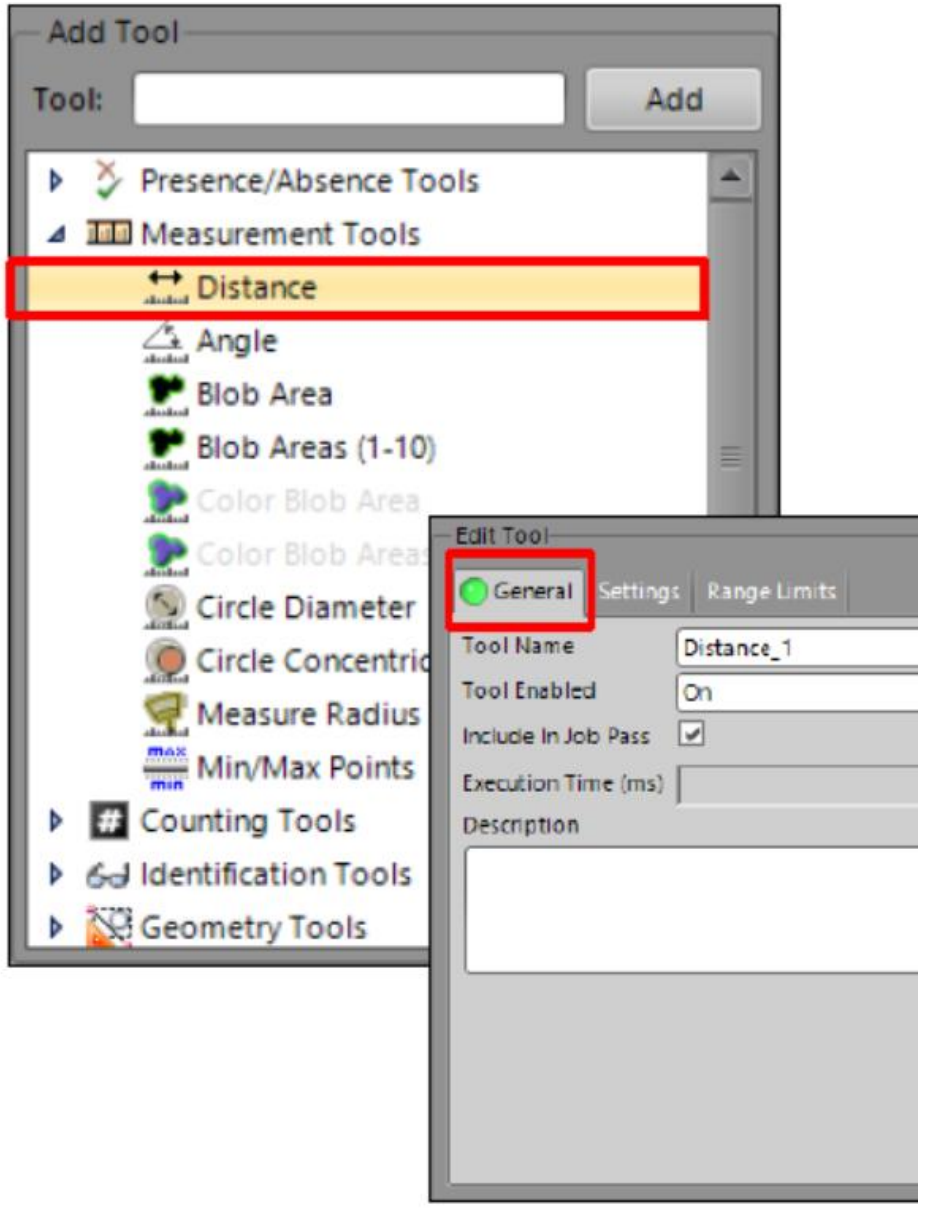
Measurement Tools

- **Measurement Tools** are used to measure the **distances, diameters, angles, and areas** of features in an image.
- Measurement Tools provide **quantitative measurements** in **pixels** or **real-world units** when the image is calibrated to real-world units in **Set Up Image**.



Distance Tool

- The **Distance Tool** measures the **distance between any two features**, such as **edges, circles, patterns, and/or blobs**. It reports the measured distance and returns a **Pass** if the measurement is within the specified limits and a **Fail** if it is outside the limits.
- The **Distance Tool** is useful when you need to **accurately gauge a part or object**. For example, it can be used to determine the **distance between the ends of a part** or the **center-to-center distance between two holes**.



Distance Tool





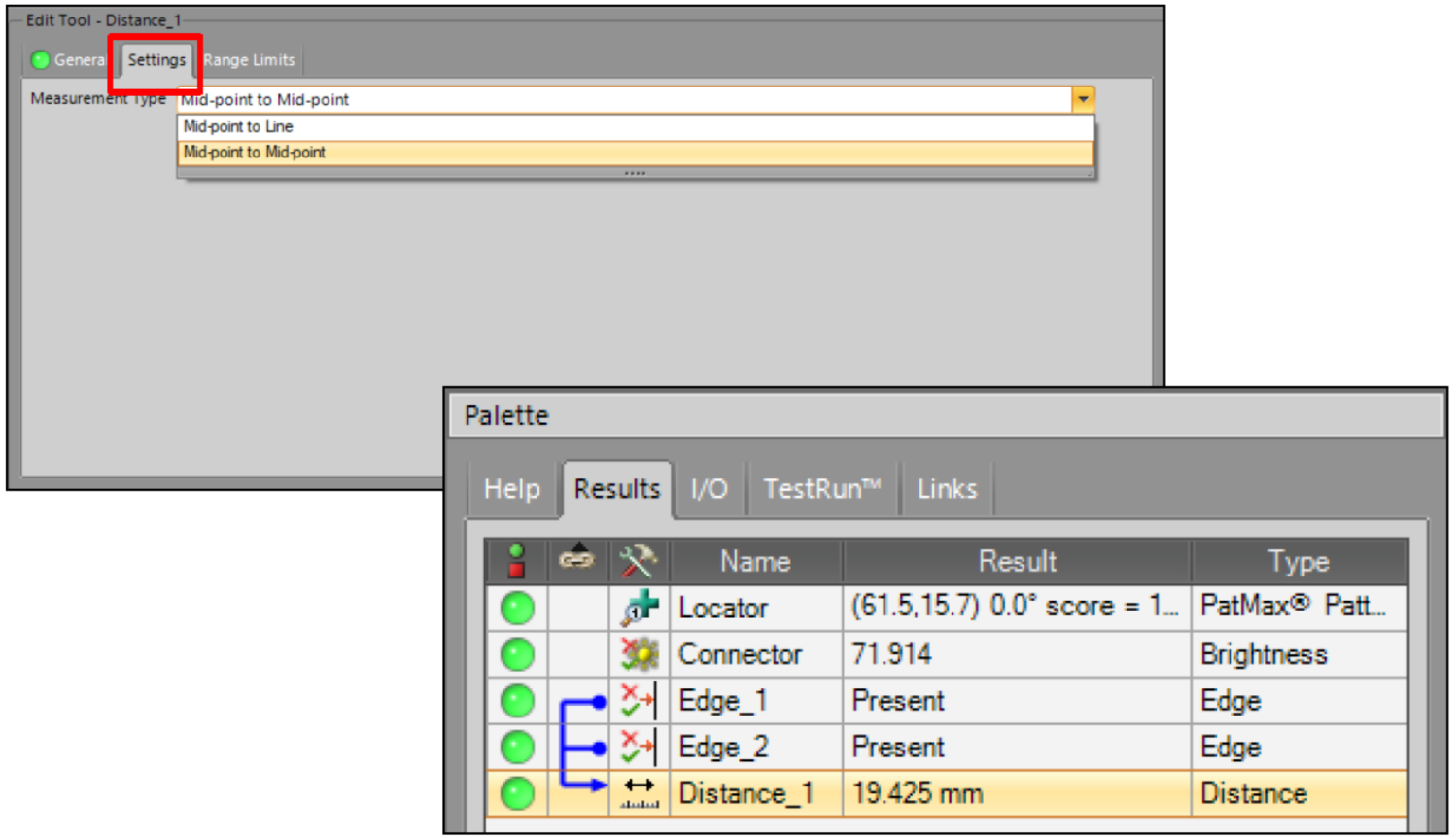


- Certain **Location and Inspection Tools** automatically extract relevant features from the image, allowing you to **point and click** to select the appropriate features for your inspection.

- If a feature has not been previously defined, it is highlighted in **light blue**, while existing features, such as **patterns, blobs, edges, or circles** defined by other Location or Inspection Tools, are highlighted in **green**.

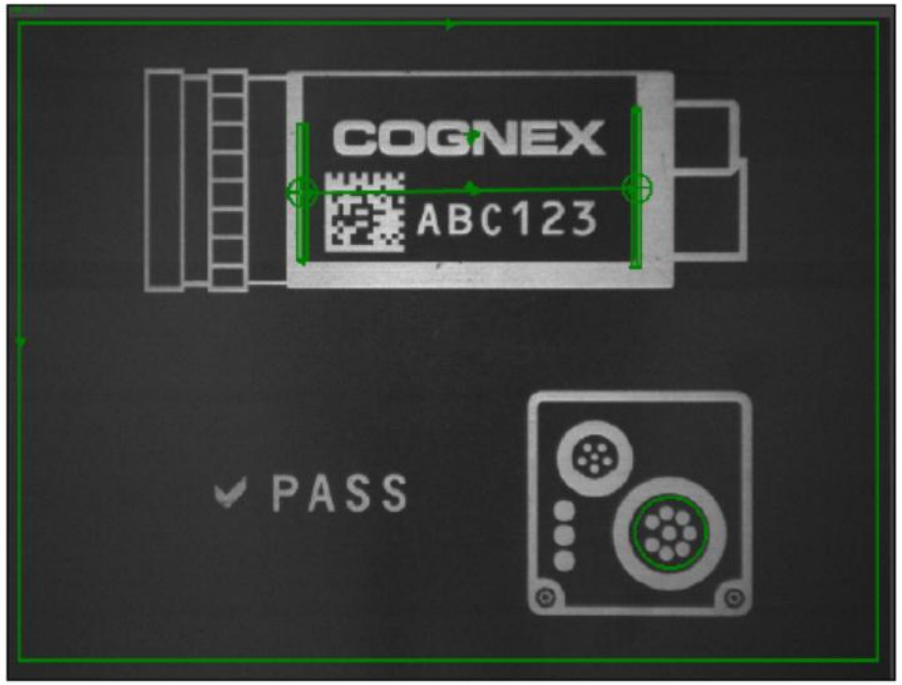
Distance Tool

- The **Distance Tool's Settings tab** specifies how the distance is calculated between two edge features:
- **Mid-point to Line** – Measures the **perpendicular distance from the midpoint of one edge segment to the other edge segment**.
- **Mid-point to Mid-point** – Measures the distance between the **midpoints of the two edge segments**.



Distance Tool - Example

- The measured distance in this image falls **within the specified limits**, so the job **Passes**.



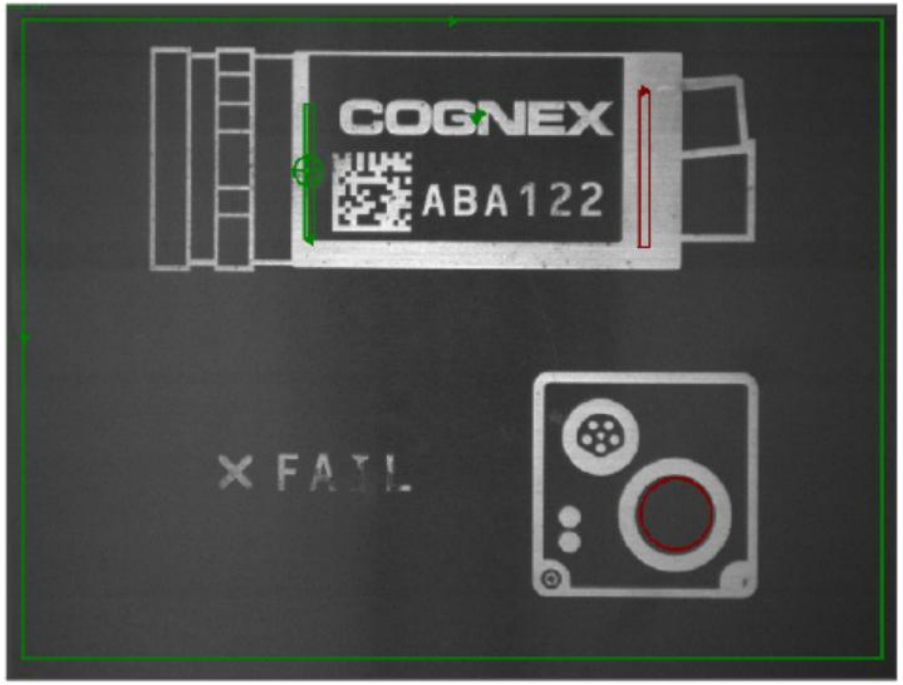
Palette

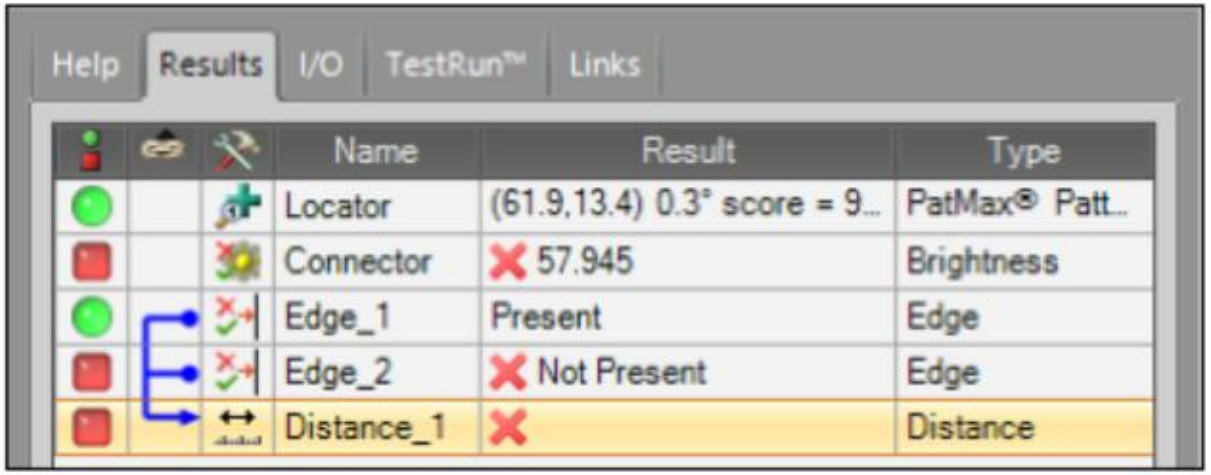
Help | Results | I/O | TestRun™ | Links

Name	Result	Type
Locator	(61.5,15.7) 0.0° score = 1...	PatMax® Patt...
Connector	71.914	Brightness
Edge_1	Present	Edge
Edge_2	Present	Edge
Distance_1	19.425 mm	Distance

Distance Tool - Example

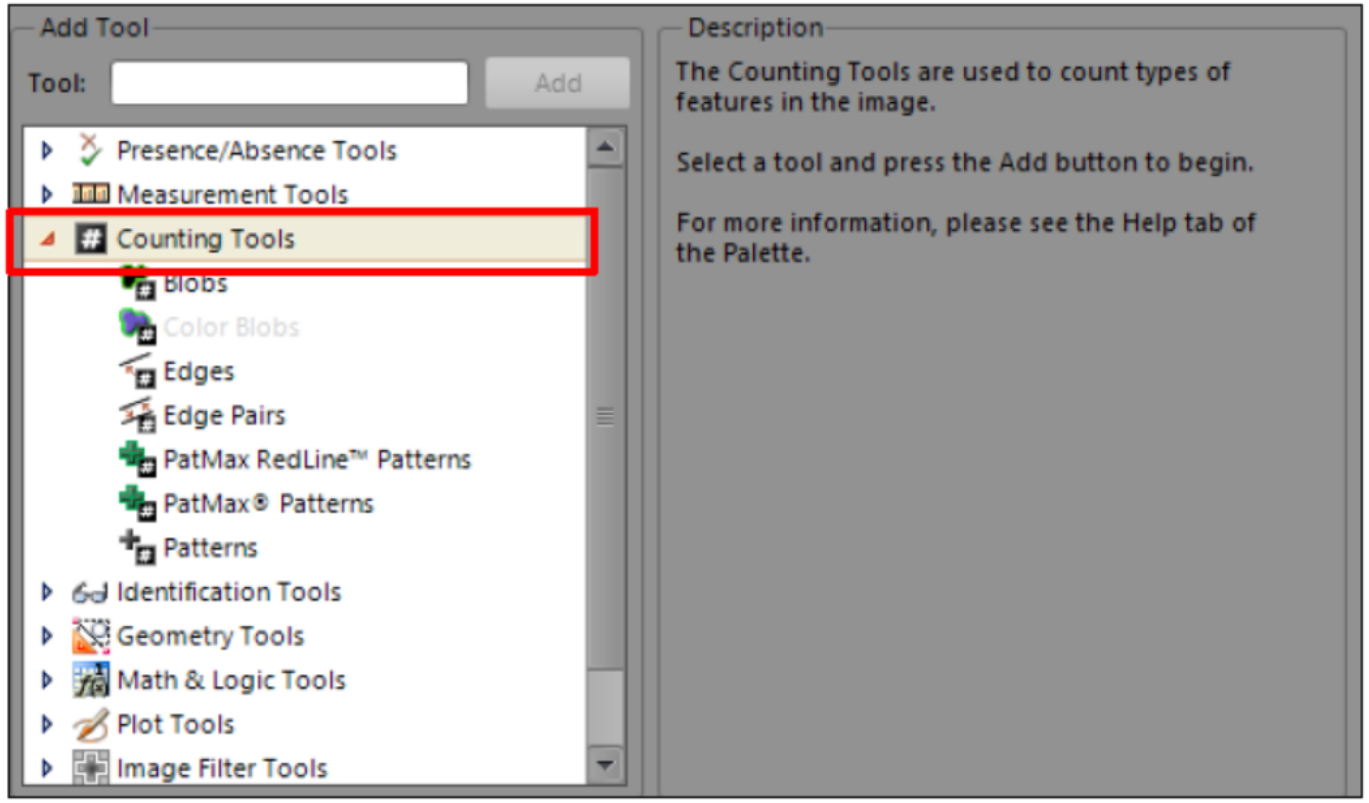
- The measured distance in this image falls **outside the specified limits**, so the job **Fails**.



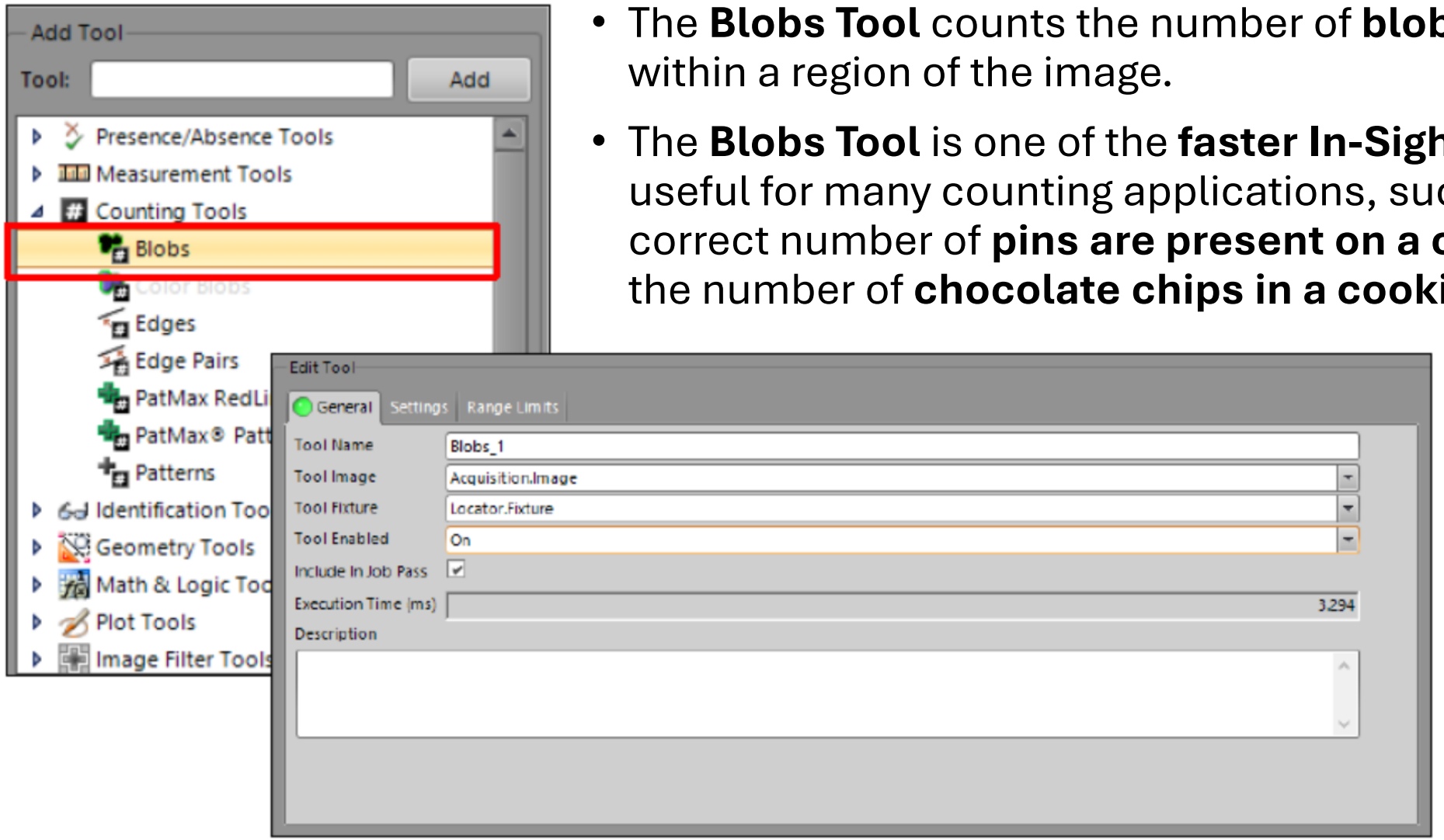


Counting Tools

- **Counting Tools** are used to **count the number of instances of a specific feature** in an image.



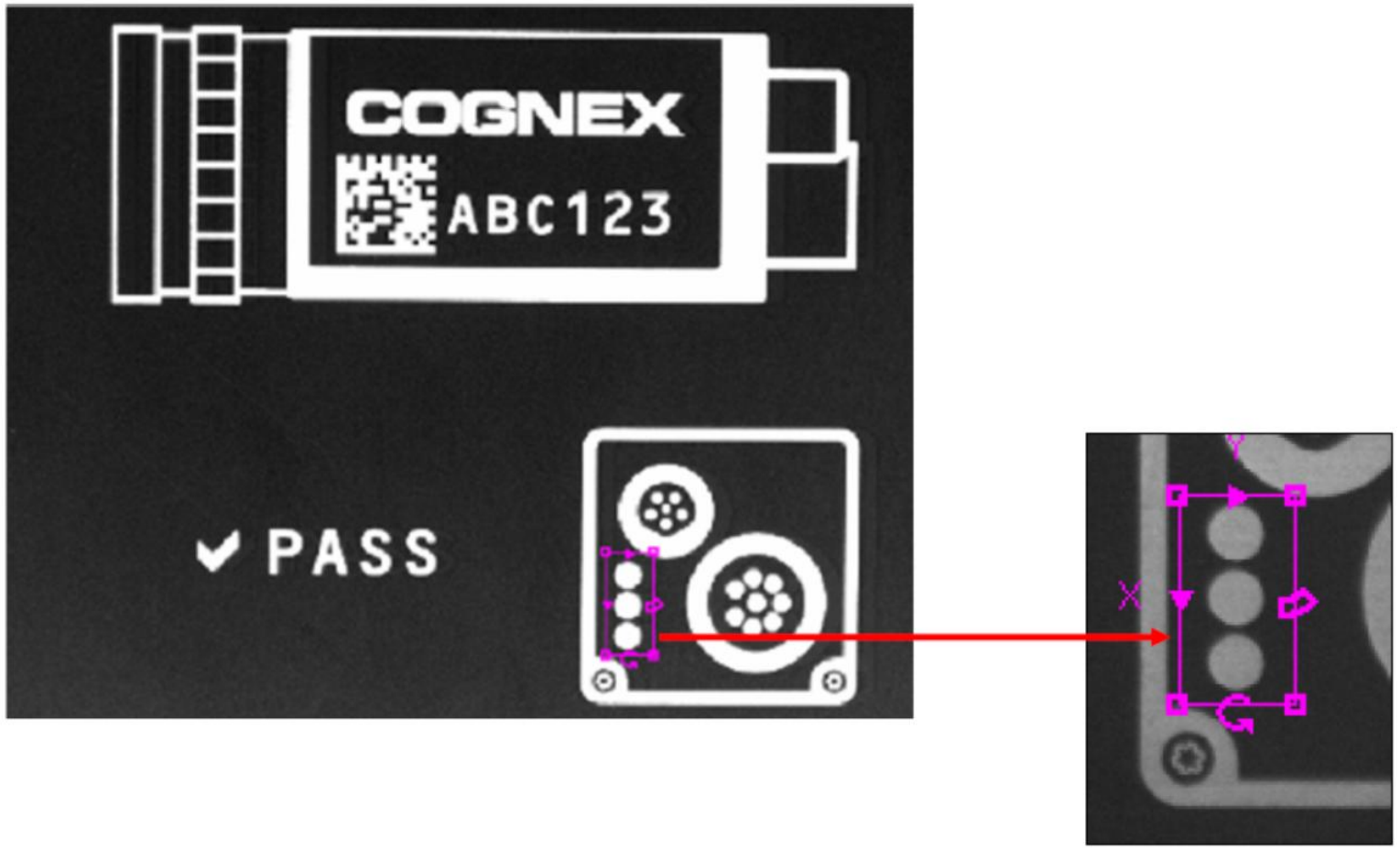
Blobs Tool



- The **Blobs Tool** counts the number of **blob features** present within a region of the image.
- The **Blobs Tool** is one of the **faster In-Sight Vision Tools** and is useful for many counting applications, such as verifying that the correct number of **pins are present on a connector** or counting the number of **chocolate chips in a cookie**.

Blobs Tool - Position Region

- Position the **region** directly over the **features you want to count**.



Blobs Tool - Parameters



Blobs Tool - Parameters

The **Blobs Tool's Settings tab** contains the following information:

- **Threshold Mode** – Defines how the **greyscale value** used to separate blobs from the background is determined. When **Auto** is selected (default), the tool automatically adjusts the threshold for changes in brightness each time an image is acquired. When **Manual** is selected, you can manually enter and adjust the threshold value.
- **Blob Threshold** – Defines the **greyscale value** used to separate blobs from the background (**0–255**). This control is **Off by default** and is enabled only when **Threshold Mode** is set to **Manual**.

Blobs Tool - Parameters

The **Blobs Tool's Settings tab** contains the following information:

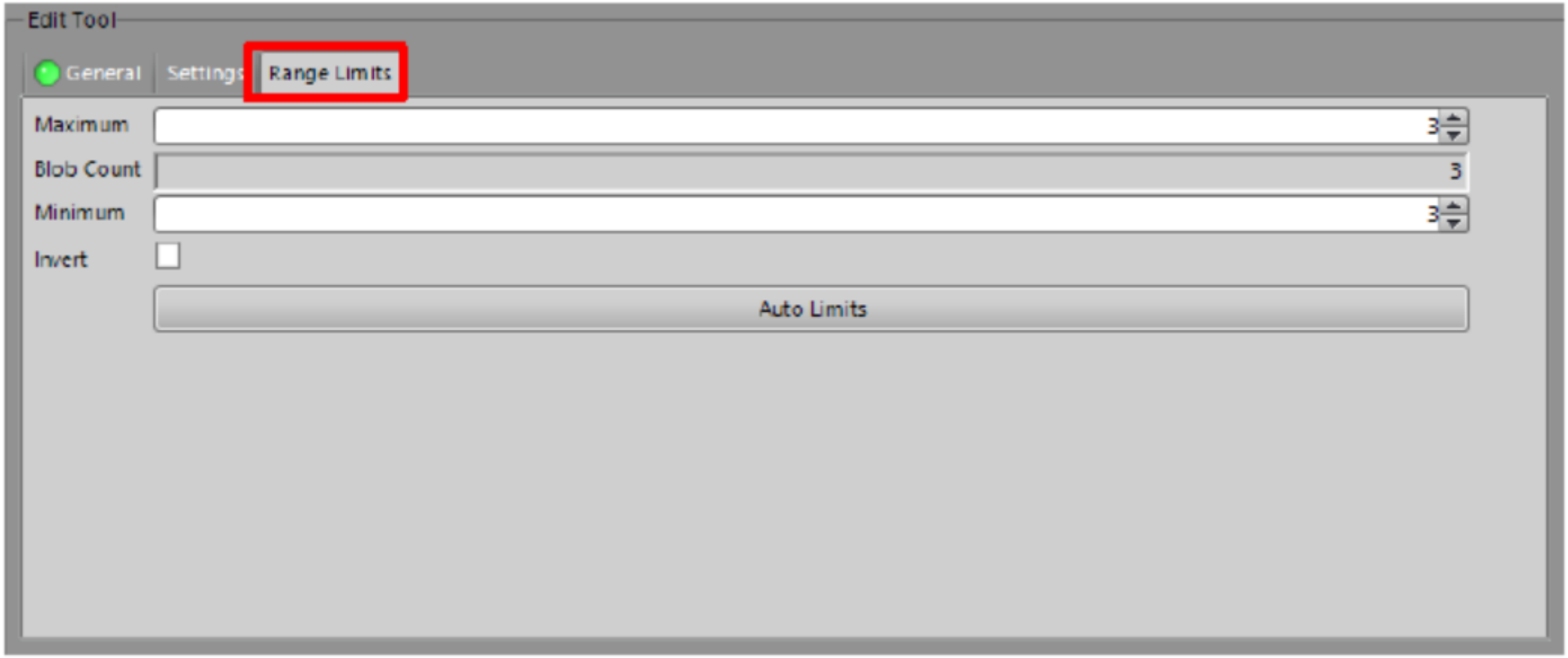
- **Blob Color** – Defines the color of the blob (**Black, White, or Either**; default = **Either**).
  - **Either –** There is no blob color requirement.
  - **Black –** Only returns blobs that are **below the Blob Threshold**.
  - **White –** Only returns blobs that are **above the Blob Threshold**.

Blobs Tool - Parameters

The **Blobs Tool's Settings tab** contains the following information:

- **Boundary** – Defines how the tool handles blobs that intersect the **region boundary** (default = **Off**, unchecked).
  - **Unchecked** – Blobs that intersect the region boundary are **not included**. The entire blob must be located within the region to be detected.
  - **Checked** – Blobs that intersect the region boundary are **included**, provided they meet the other specified criteria.
- **Minimum / Maximum Area** – Defines the **minimum and maximum blob area**, in pixels (**0–900,000**). Only blobs with an area **greater than or equal to the Minimum Area and less than or equal to the Maximum Area** are recognized.

Blobs Tool - Range Limits



Blobs Tool - Range Limits

The **Blobs Tool's Range Limits** tab contains the following information:

- **Maximum / Minimum** – Defines the **maximum and minimum acceptable number of blobs** to detect. The tool creates default **Smart Limits** based on the image, which you can customize to suit your application.
- **Blob Count** – Reports the **current number of detected blobs**. The blobs are counted using a numerical order where **0 = 1**. Values that fall **at or between the Minimum and Maximum settings** are classified as **Pass**.

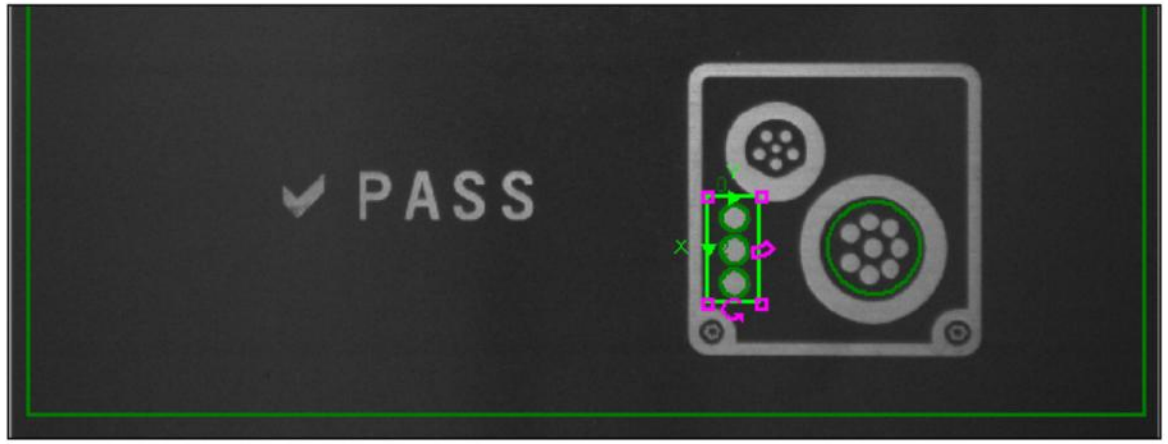
Blobs Tool - Range Limits

The **Blobs Tool's Range Limits** tab contains the following information:

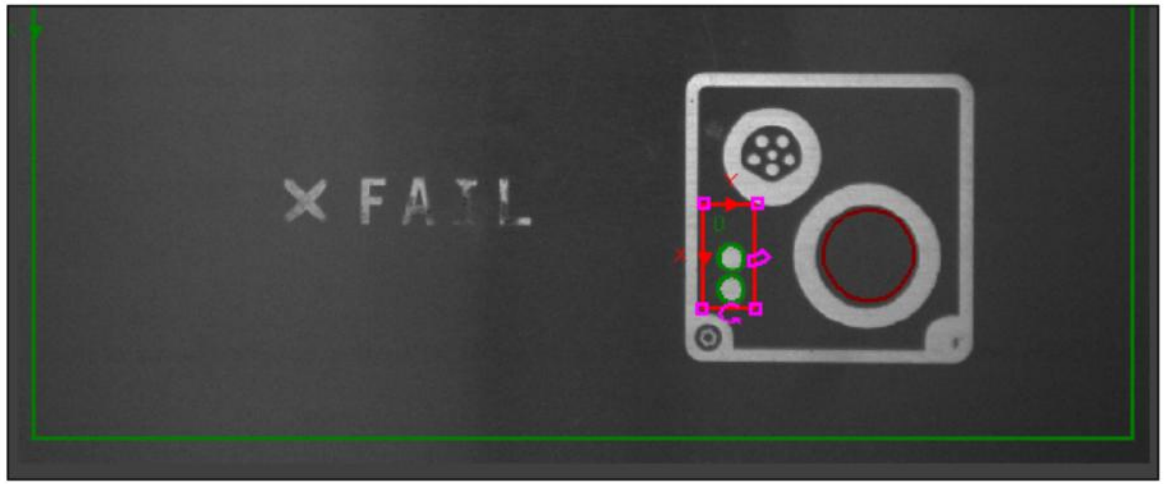
- **Invert Result –** Defines whether the **Pass/Fail logic** of the tool should be inverted (default = **Off**, unchecked).
  - **Unchecked –** Standard results are applied. The tool result is **Pass** if the feature is detected and falls within the specified settings.
  - **Checked –** The results are inverted. The tool result is **Fail** if the feature is detected and falls within the specified settings.
- **Auto Limits Button –** Defines the limits based on the **Smart Limits** settings. This button is useful when a parameter has been changed and the limits need to be adjusted, or when a tool has been added and no limits have been defined.
- **CAUTION**: *Depending on the lighting and image resolution, blobs can touch or overlap, causing them to be detected as a single, larger blob.*

Blobs Tool

- The first image **detects three blobs**, so the job **Passes**.
- The second image **detects only two blobs**, so the job **Fails**.



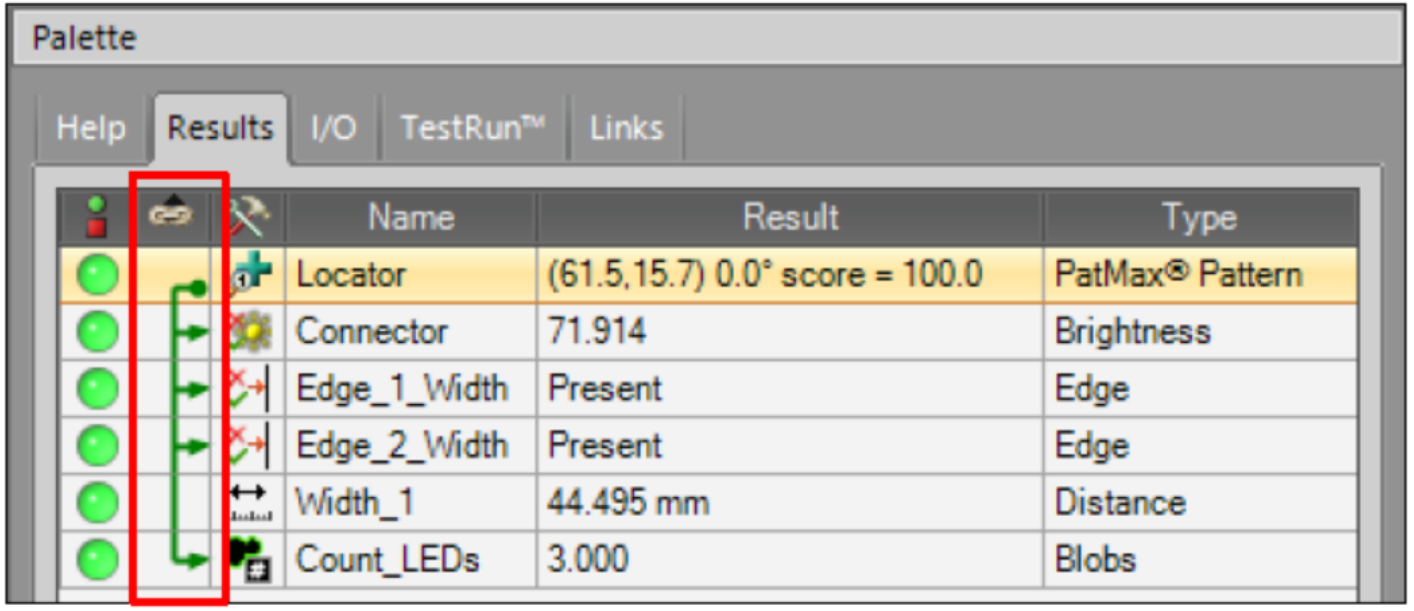
3 Blobs = Pass



2 Blobs = Fail

Palette and Dependencies

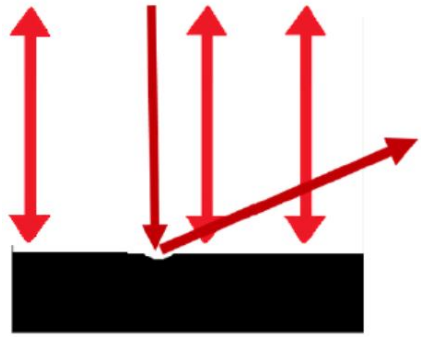
- The arrows in the **second column** indicate **dependencies**, showing which tools depend on other tools.
- In this example, **five tools** depend on the **Location Tool** because they are **fixtured to it**.



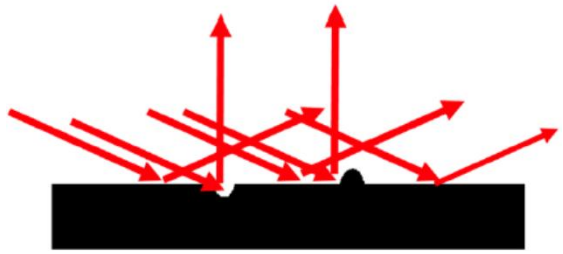
Lighting

Lighting

- **Lighting** is the way light illuminates and interacts with a part to create an image
- **Proper lighting** is important in machine vision because it can make important **features easier to detect** and **reduce unwanted features or variations**.
- One way to describe lighting is based on **how the part and its features appear in the image**.
- Two basic lighting effects are **bright field** and **dark field**.



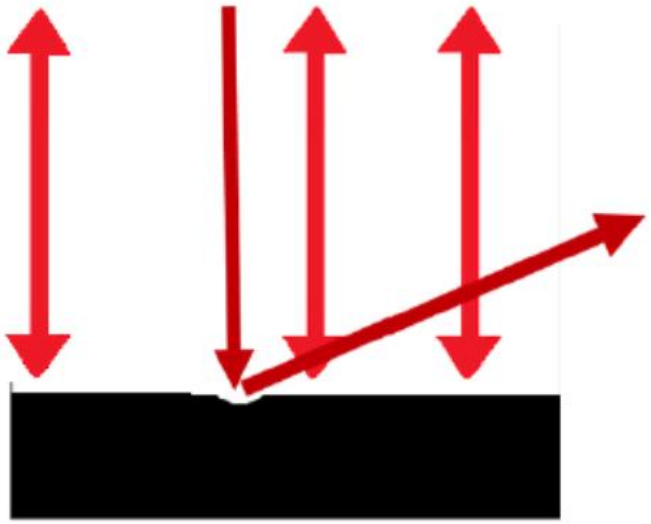






Bright Field

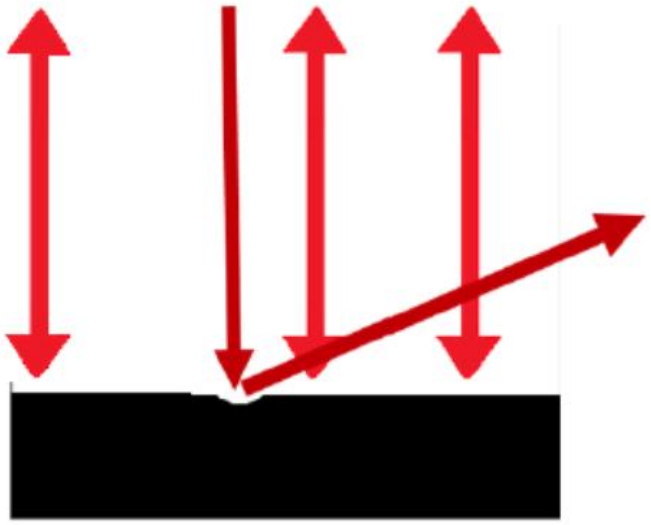
- **Bright field lighting** illuminates the part so that light reflected from relatively flat surfaces is directed back toward the camera. As a result, these surfaces appear **bright** in the image.
- **Indentations or raised features** change the direction of the reflected light. Some of the light is reflected away from the camera, causing these features or their edges to appear **dark**, which helps reveal their shape.





Bright Field

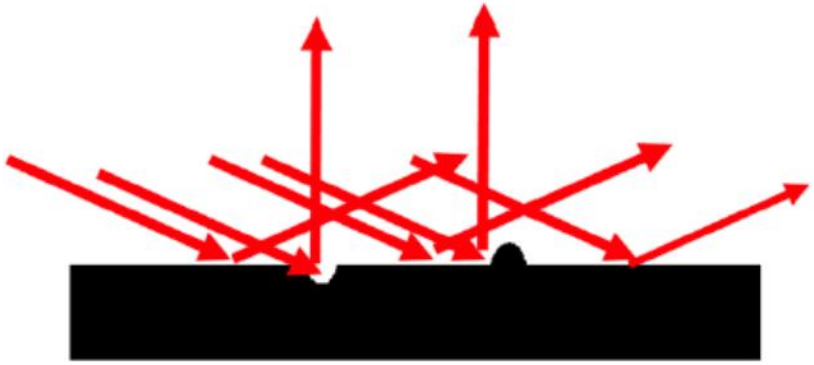
- Looking at this coin, when the **light rays are perpendicular to the surface**, the flat surfaces reflect light back toward the camera, creating **bright areas**, while the indentations or raised features reflect light away from the camera, appearing **dark**.
- The **shape and contours** of the coin are enhanced.
- **Bright-field lighting can be used to emphasize surface features and height changes.**





Dark Field

- **Dark-field lighting** works in the opposite way to bright-field lighting.
- The **light strikes the part at a shallow angle**.
- **Flat surfaces** reflect most of the light away from the camera, appearing **dark**, while **raised or indented features** can redirect light toward the camera, making them appear **bright**.
- The **shape and contours** of surface features are enhanced.
- **Dark-field lighting is used to emphasize surface features and height changes.**



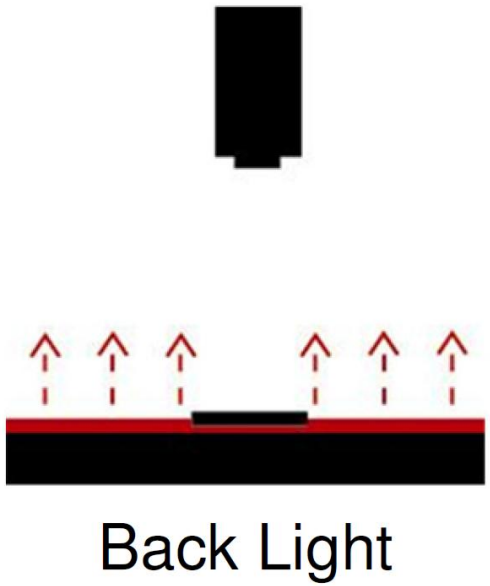


Lighting Techniques

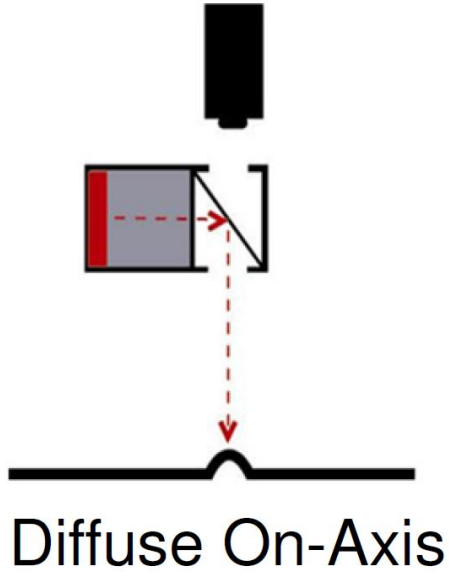
- **Bright field and dark field describe the appearance of the image.**
- To produce these lighting effects, we can use different **lighting techniques**, depending on the position and type of the light source.
- A lighting technique is a combination of a **light source** and how it is **positioned relative to the part and the camera**.
- Different lighting techniques change how light reaches the part and how reflected light reaches the camera.

Lighting Techniques

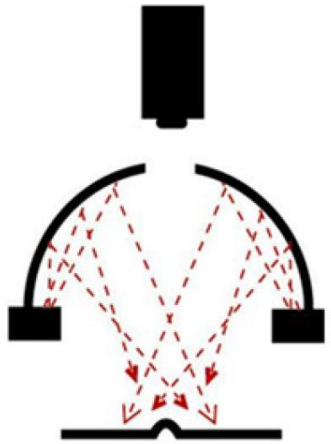
- There are five basic lighting techniques: **Direct**, **Back**, **Diffuse On-Axis**, **Diffuse Off-Axis**, and **Structured**



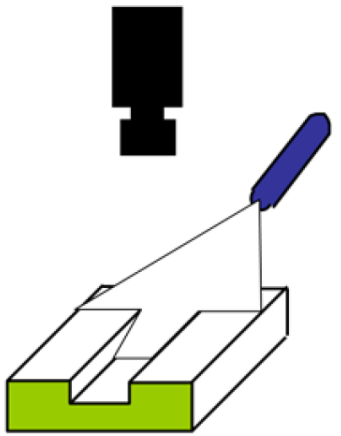
Back Light



Diffuse On-Axis



Diffuse Off-Axis



Structured

Direct Light

- **Direct light** is a basic lighting technique in machine vision where the light source shines **directly onto the part and toward the camera**.
- Here's what happens:
  - Light hits the surface **directly**.
  - **Smooth, reflective surfaces** reflect the light directly back toward the camera, appearing **bright**.
  - **Rough or angled surfaces** scatter the light away from the camera, appearing **darker**.



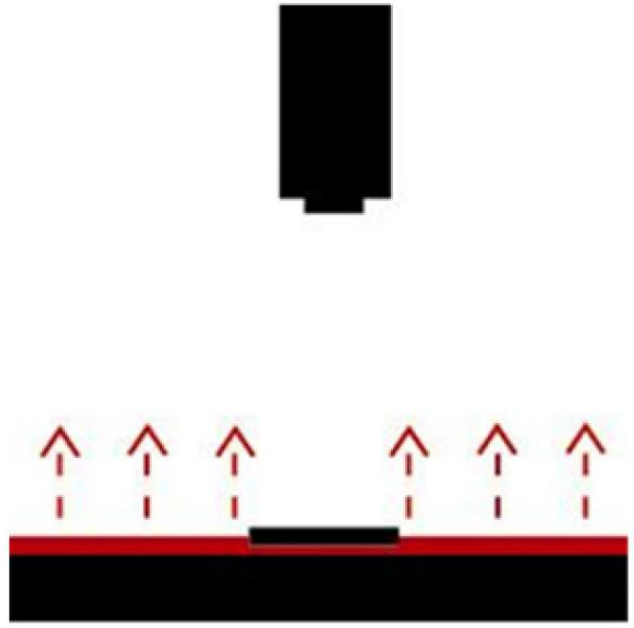
Direct Light

- **Direct light** is useful for highlighting **flat, reflective surfaces** or **surface defects**.
- However, it can create **glare or hotspots** on shiny parts, which is why other lighting techniques are often used instead.



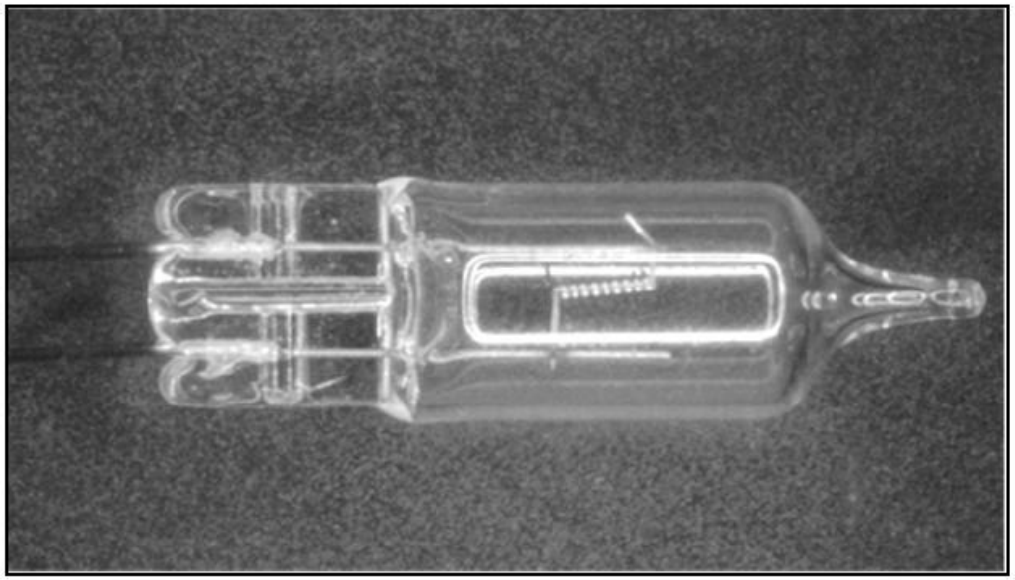
Back Light

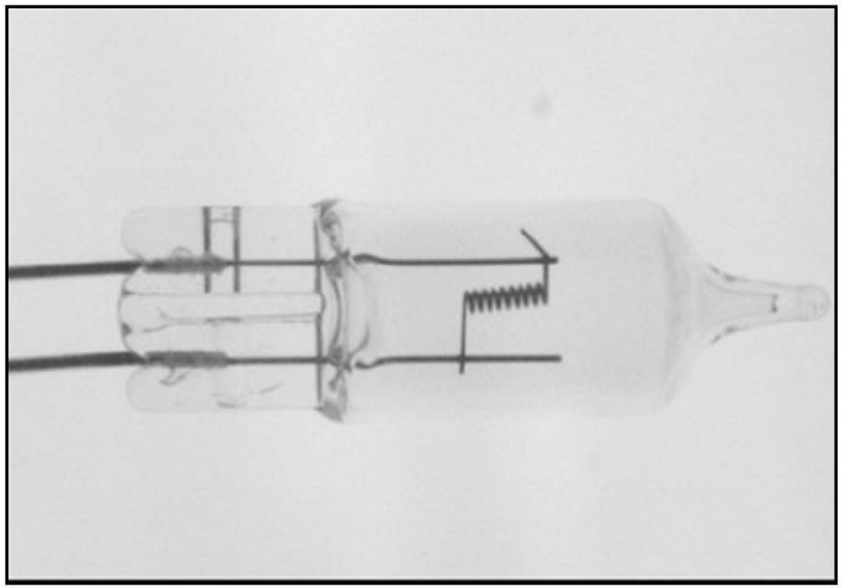
- **Back light** creates a **silhouette** that highlights the shape of a part for measurement while removing surface details.
- It provides **strong contrast**, typically producing a **dark part against a bright background**.



Back Light Application

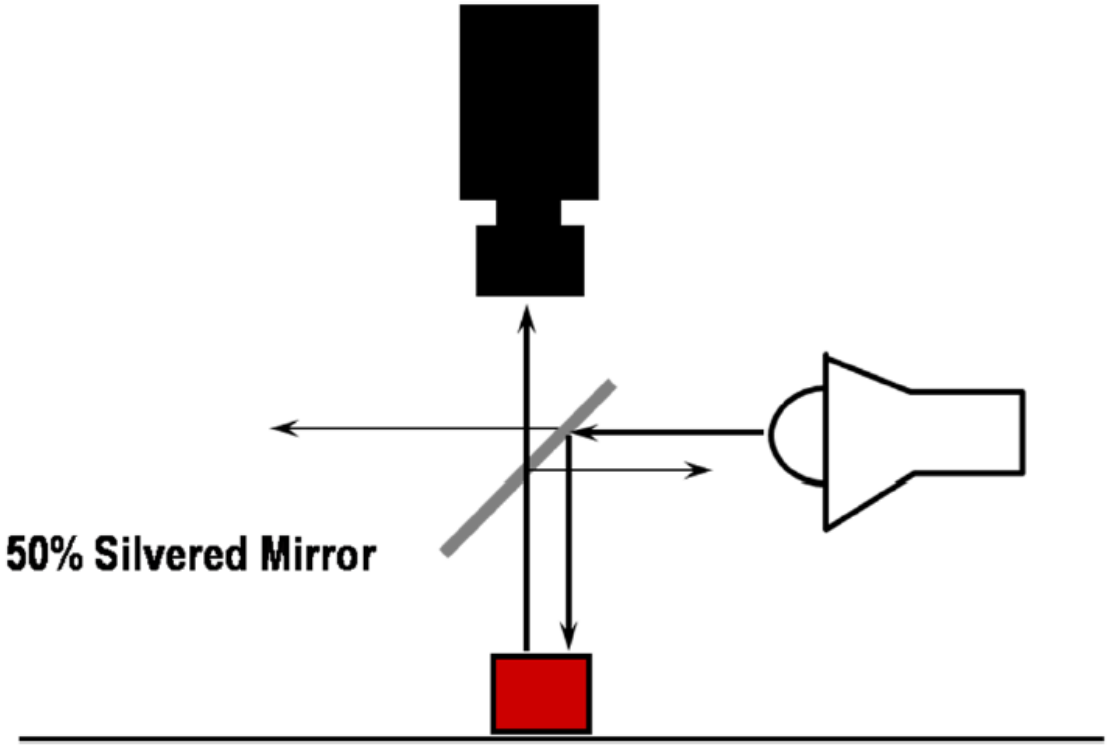
- This application measures the **filament of a light bulb**.
- In the **top image**, the light bulb is illuminated using **direct light**. Note the **glare from the glass**, which makes it difficult to clearly see the bulb's filaments.
- In the **bottom image**, using **back lighting** makes the **filaments inside the bulb** much easier to see.





Diffuse On-Axis Light

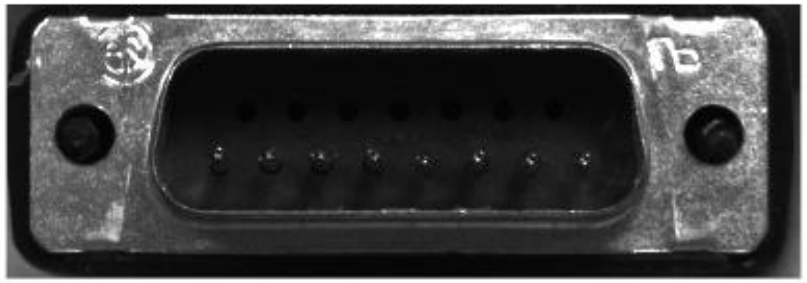
- **Diffuse On-Axis Lighting** directs light **straight onto the part** without blocking the camera's view.
- This is achieved using a **half-silvered (50%) mirror**: the light shines onto the mirror and is reflected down onto the part, while the camera looks **through the mirror** to capture the illuminated image.

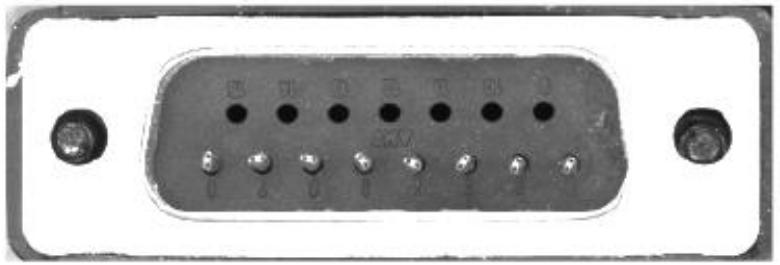


A 50% silvered mirror is a mirror that reflects approximately half of the light that strikes it and transmits the other half

Diffuse On-Axis Light Application

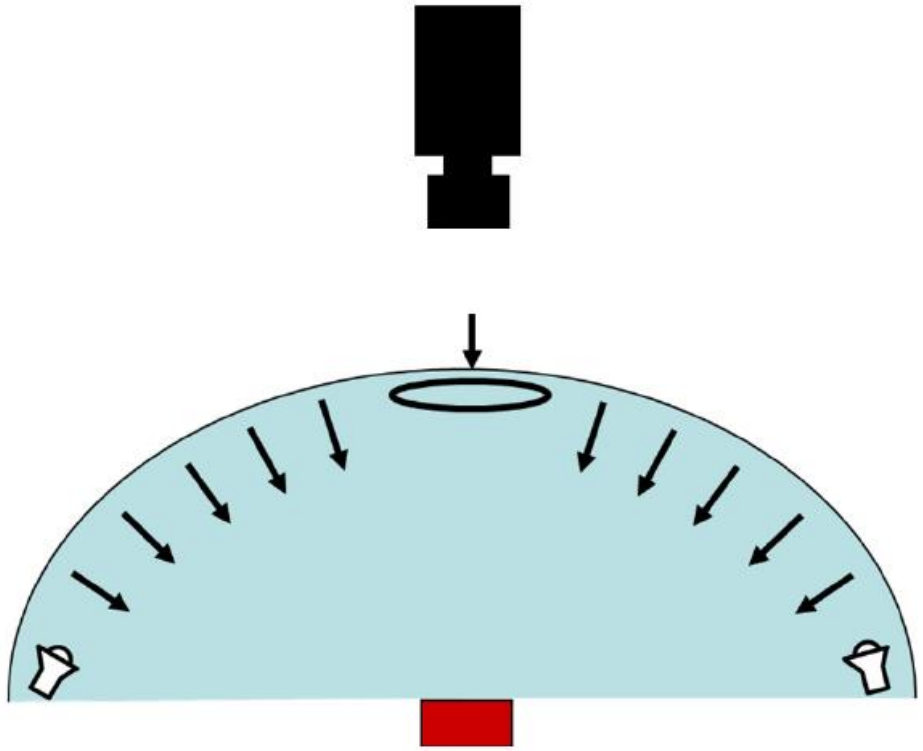
- Here is a **dot-pinned connector** with a rough surface. The image on the **left** shows the connector under **ambient light**.
- The image on the **right** shows the connector under **Diffuse On-Axis Lighting**. Notice how the **dots are accentuated** while the **rough background is suppressed**. There are also **no hotspots or glare**.





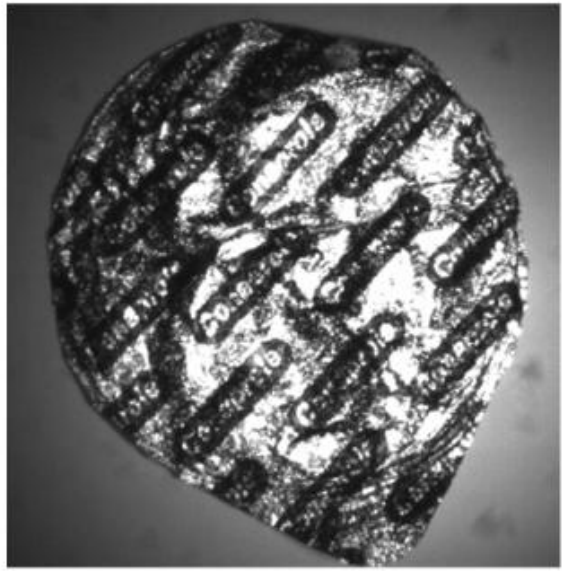
Diffuse Off-Axis Light

- With **Diffuse Off-Axis Lighting**, the light does not shine directly onto the part. Instead, it first strikes a **diffuse surface**, such as the inside of a dome, which scatters the light evenly. The scattered light then illuminates the part.
- In the image, the **lights at the bottom of the dome** shine upward, and the **dome scatters the light back down** onto the part, creating **soft, uniform illumination**.



Diffuse Off-Axis Light Application

- Here is a **foil lid inspection** example. Under **direct lighting**, the shiny surface produces strong **glare**, making defects difficult to see.
- With **Diffuse Off-Axis Lighting**, the glare is reduced, allowing the **entire lid** to be inspected for **holes or tears**.



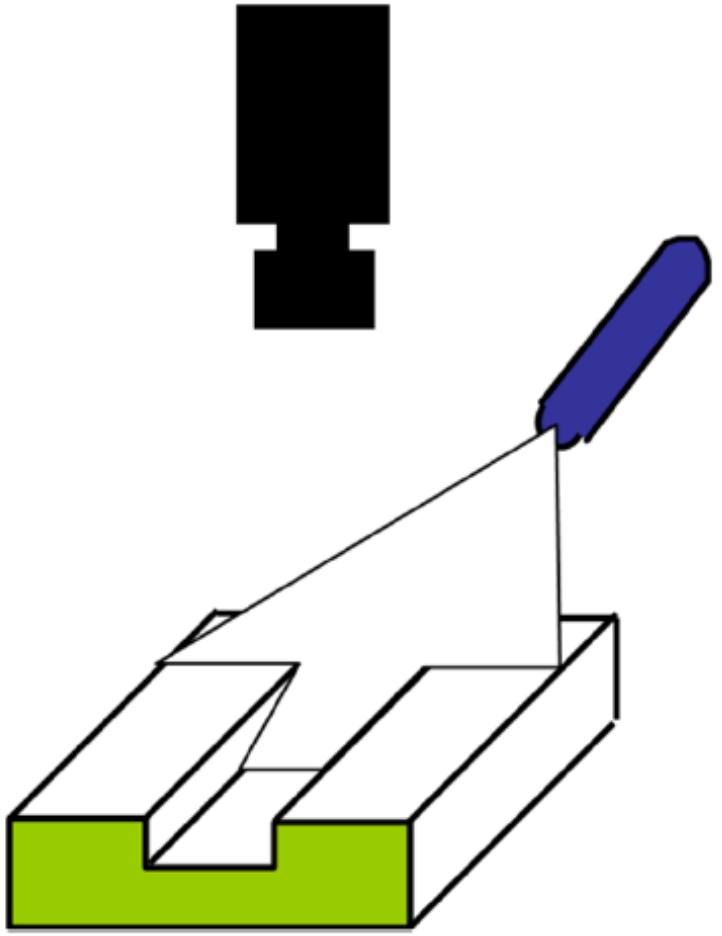
Direct Light



Diffuse Off-Axis Light

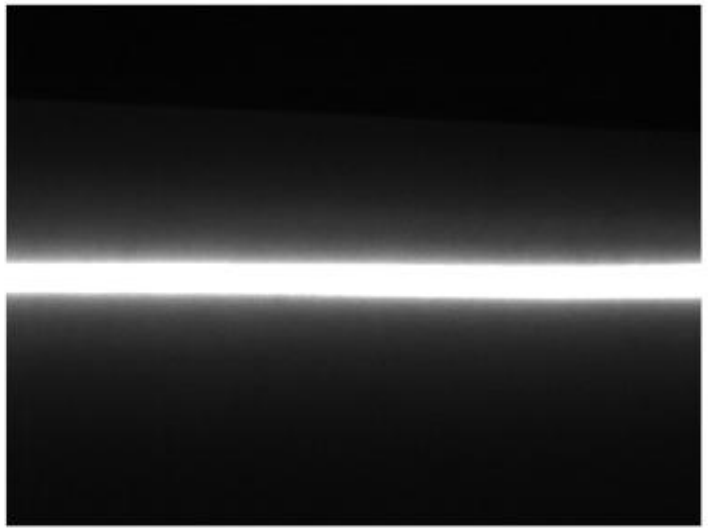
Structured Light

- The last technique is called **Structured Lighting**.
- It uses a **known light pattern**, usually a **plane of light forming a line**, to capture **dimensional information** about a part.
- The light sources are typically **highly collimated**, such as **lasers or fiber-optic line lights**, to maintain a **precise and sharp pattern**.

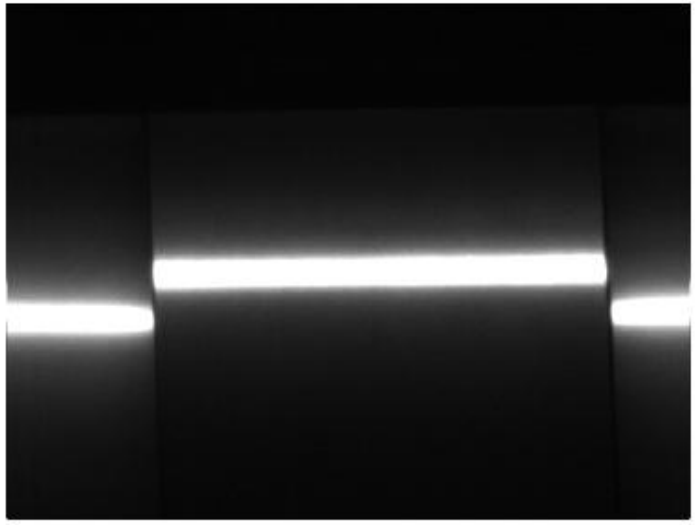


Structured Light Application

- In this application, we are inspecting **changes in the depth of a part**.
- On the **left**, the line is **flat**, indicating a flat surface.
- On the **right**, the line is **broken**, indicating a **step in the middle**. The displacement of the line corresponds to the change in **height (Z dimension)** — the larger the offset, the greater the difference in depth.



Line Light



Change in Depth

Ambient Light

- **Ambient lighting** is any external light that illuminates the parts being inspected, such as **ceiling lights or sunlight from skylights**.
- It can be a problem because it **varies over time and can affect inspection results**.



Ambient Light

Ways to deal with ambient lighting:

- **Eliminate it** – Cover skylights or turn off overhead lights.
- **Block it** – Enclose the parts being inspected to prevent ambient light from reaching them.
- **Filter and control your own light** – If ambient light cannot be removed or blocked, use a **color filter** and illuminate the part with a light source of the **same wavelength** (for example, red light).

